Brain Before the Swipe: Can Consumer-Grade EEG Predict Viewer Engagement during Naturalistic TikTok Browsing?

Master's Thesis

Gregor Lederer

Supervisors:
Dr. Irina-Catrinel Craciun
Prof. Dr. Dr. h. c. mult. Lothar H. Wieler
Prof. Dr. Bert Arnrich

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Hasso Plattner Institute
University of Potsdam

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Contents

Abstract

This thesis investigates whether consumer-grade Electroencephalography (EEG) can predict when a user is actively sustaining cognitive engagement during naturalistic TikTok browsing. Using a Muse S headband (4 channels, 256 Hz), we recorded brain activity from 25 participants while they freely browsed their personal TikTok feed without artificial constraints. Behavior was mapped via tracking video transitions (screen swipes), allowing us to construct a rigorous binary classification framework focused on the underlying cognitive state: *sustained engagement* (`STAY`, Class 1) versus *imminent task abandonment* (`SKIP`, Class 0), captured via the 3-second pre-skip window.

We implemented a robust pipeline applying baseline-normalized relative spectral power features. A Random Forest classifier (200 trees, 112 frequency-domain features) was evaluated under a blocked temporal 5-fold cross-validation protocol—designed to eliminate autocorrelation between train and test sets—yielding a mean intra-subject accuracy of **53.8%** (Wilcoxon $p = 0.0015$ vs. chance), with 20 of 25 participants exceeding the 50% baseline. Three participants demonstrated strong individual classification (P12: 81.2%, P7: 67.3%, P22: 62.9%).

Feature importance analysis revealed that predictive variance was distributed broadly across all seven frequency bands, with delta (17.5%) and theta (14.8%) capturing slightly more aggregate Gini importance than high-gamma (14.7%), and no single band achieving statistical dominance. At the individual feature level, peak high-gamma values at frontal sites were the top-ranked predictors, suggesting the model captures transient high-frequency volatility within a whole-brain spectral pattern. An empirical ablation of the 50 Hz notch filter showed no significant performance change (Wilcoxon $p = 0.76$), consistent with the interpretation that the 40–60 Hz range contains recoverable cognitive variance. The standard clinical Engagement Index ($\beta/(\alpha + \theta)$) failed to discriminate these states (52.3% accuracy), confirming that full structural spectral representations are required. Cross-subject generalization via LOGO-CV collapsed to chance (49.2%), confirming the idiosyncratic nature of the engagement signature.

These findings establish a proof-of-concept that consumer-grade EEG captures statistically significant, individualized cognitive engagement states during naturalistic algorithmic media consumption, while demonstrating that standard overlapping-window evaluation in BCI research can substantially inflate reported classification performance. Generalization to broader populations remains an open question requiring demographically diverse replication; the present contribution is the feasibility demonstration and methodological critique itself.

Chapter 1

Introduction

Short-form video platforms like TikTok now fundamentally shape how billions of people allocate their attention, often in bursts lasting mere seconds. On these platforms, every single swipe represents a critical micro-decision: to stay and engage, or to skip and move on. Repeated hundreds of times per session, these sequential micro-decisions determine exactly what information people absorb, which health or educational messages effectively reach them, and what digital behaviors are continuously reinforced.

For public health communication, this dynamic poses a significant challenge. Health promotion interventions delivered via social media must compete head-to-head with highly optimized entertainment content for the exact same limited cognitive resources [1]. Systematic review evidence confirms that TikTok has been adopted for a range of public health purposes, yet institutional accounts remain poorly engaged relative to personal creators precisely because they fail to leverage the platform's full attentional affordances [1]. If a health promotion video fails to maintain engagement in the critical first few seconds, the core message never lands. To improve these interventions, we need to understand what actually causes a human brain to sustain engagement—not just what individuals retrospectively claim they liked, but rather what their objective neural state looked like during active, continuous viewing. This investigation will not help us determine wether a specific content-piece changes behaviour and why, but rather we investigate neural clues for potential behaviour change and other neural markers that may be psychiologically and behaviourally relevant for optimized behaviour change messaging in public health and beyond.

Understanding the neural signature of sustained engagement—and the exact moment that engagement begins to collapse—opens the door to designing communications that maintain attention precisely when the brain is most vulnerable to disengagement. This investigation is about understanding how to make health-relevant content as neurologically compelling as the content it competes against.

Specifically, this thesis asks: within a controlled proof-of-concept study, can consumer-grade EEG—using a 4-channel dry-electrode headband—reliably classify sustained cognitive engagement versus imminent disengagement during naturalistic TikTok browsing at the intra-subject level? As a secondary question, and with explicit acknowledgement of the sample's demographic homogeneity, this thesis further examines whether any such intra-subject neural signature shows preliminary evidence of cross-individual consistency.

Chapter 2

Literature and State of the Art

The following sections outline the core theoretical claims, methodologies, and prior research required to contextualize this study.

2.1 Public Health, Media Consumption, and The Attention Economy

- **Claim:** Short-form video algorithms (e.g., TikTok) fundamentally condition human attention into rapid micro-bursts and continuous micro-decision making. Empirical evidence for this attentional fragmentation comes from Baughan et al., who deployed a custom social media client to 43 participants and found that passive feed browsing induces a state of *normative dissociation*—total cognitive absorption characterized by diminished self-awareness and reduced sense of agency—in which users describe mindlessly scrolling while "not even remember[ing] what they read" [2]. Crucially, this dissociative state is not equivalent to genuine engagement: participants described passively slipping into it without intention, failing to absorb any content, and feeling as though they had wasted their time. The study further demonstrated that this state is actively engineered by platform design features including infinite scroll and variable reward schedules, which "harn[ess] people's natural inclination to seek normative dissociation" [2]. Every swipe in the present thesis therefore represents not a conscious content appraisal but the behavioral residue of a fragmented attentional micro-decision occurring within this dissociative context. This dissociative baseline does not argue against health communication on TikTok — it sharpens the research imperative: because re-engagement events do occur episodically within the dissociative stream [2],identifying the neural signature that

reliably precedes attentional lock-in becomes the precise design lever for health content that crosses the threshold.
Search Query: `"short-form video" OR TikTok "sustained attention" OR "attention span"`

- **Claim:** Public health and educational communication struggles to survive on social media because it competes against highly optimized entertainment for the same limited cognitive resources. McCashin and Murphy's systematic review of TikTok for public and youth mental health found that while the platform has been deployed across a wide range of health topics, institutional accounts consistently underperform personal creators in engagement metrics, with personal creators generating orders-of-magnitude higher reach precisely through fuller exploitation of the platform's algorithmic and audiovisual features [1].
Search Query: `"public health communication" "social media" "cognitive load" OR "attention economy"`

- **Claim:** Continuous social media feeds are designed to bypass conscious appraisal and induce a state of behavioral momentum, creating a high cognitive barrier for educational interventions. Knierim et al. found that passive feed consumption—the dominant form of social media engagement—produces a "pseudo-flow" state characterized by absorption and time distortion without the goal-directed engagement of authentic flow, noting that passive browsing "often lacks clear goals and meaningful challenge" and can produce dissociative experiences where users scarcely register the content they have browsed [3]. This pseudo-absorptive state is not driven by conscious content appraisal but by the structural properties of the feed itself—precisely the condition that makes deliberate engagement with health content neurologically costly. Baughan et al. provide the complementary behavioral phenomenology: users passively slip into normative dissociation during scrolling such that their self-reflection and self-awareness is suspended, and because dissociation by definition involves a diminished sense of volition, users cannot simply self-regulate their way out of the scrolling loop [2]. This is the precise cognitive barrier that makes delivering health-relevant content inside an algorithmic feed structurally difficult.
Search Query: `"flow state" OR "behavioral momentum" "social media" cognitive barrier intervention`

- **Claim:** Retrospective self-report is structurally insufficient for capturing the micro-decision dynamics described above. Because normative dissociation by definition suspends self-awareness [2], and pseudo-flow states produce time distortion and reduced content registration [3], participants cannot reliably reconstruct which specific moments held their attention and which did not. This is not a general indictment of self-report methodology but a specific consequence of the attentional states this thesis targets: you cannot accurately report on a cognitive state that, by its nature, you were not consciously tracking. Knierim et al. explicitly identify this limitation and propose that physiological signals reflecting attentional dynamics, such as EEG, serve as objective indicators to empirically separate genuine engagement from pseudo-flow states during social media use [3]. The present thesis operationalizes exactly this recommendation, replacing post-hoc self-report with continuous EEG classification of the `STAY` versus `SKIP` cognitive state. *Search Query:* `"neural correlates" media engagement "self-report bias" OR "retrospective"`

2.2 Neuroscience of Sustained Attention vs. Micro-Decisions

- **Claim:** In educational and public health contexts, evaluating engagement requires prioritizing the detection of sustained attentional maintenance (the "STAY" state) over the mere detection of attentional collapse, as maintenance is the prerequisite for memory encoding. Trial-by-trial fluctuations of sustained attention have been shown to directly predict which items are subsequently remembered, with stronger sustained attention correlating with better long-term memory performance across individuals [4].
  Search Query: `"sustained attention" maintenance "memory encoding" OR learning EEG`

- **Claim:** The classical EEG Engagement Index ($\beta/(\alpha + \theta)$) was designed for sustained vigilance (e.g., automated flight-deck monitoring) and is fundamentally too coarse to capture rapid algorithmic micro-decisions [5]. Critically, Pope et al. validated the formula exclusively under slow-changing, controlled task-demand conditions; the present study deliberately applies it to a high-speed naturalistic paradigm precisely to expose this boundary.
  Search Query: `"EEG Engagement Index" "beta / (alpha + theta)" vigilance "sustained attention"`

- **Claim:** Alpha band activity reflects cortical idling — regions that are neither processing sensory input nor generating motor output [6]. When a cortical area has "nothing to do," its neurons synchronize into a characteristic alpha rhythm; when that area is recruited for active processing, this rhythm desynchronizes and alpha power drops. This has a direct structural consequence for the Engagement Index: if alpha power in the denominator tracks passivity rather than a fixed physiological baseline, the ratio $\beta/(\alpha+\theta)$ becomes unstable under rapid, dynamically shifting attentional states. In a paradigm where participants transition between engagement and disengagement within seconds, the alpha denominator cannot be assumed to behave classically — which predicts, before any data is collected, that the EI will struggle to discriminate `STAY` from `SKIP` in this context. *Search Query:* `"EEG alpha band" "cortical idling" OR relaxation "decision making"`

- **Claim:** Active cognitive engagement, complex binding, and algorithmic content appraisal are heavily mediated by high-frequency Beta and Gamma bands.
  Search Query: `"EEG beta" OR "high gamma" "sustained cognitive engagement" OR "active appraisal"`

- **Claim:** The prefrontal cortex (AF7, AF8) serves as the primary cortical substrate for top-down cognitive control, maintaining goal-relevant representations and biasing downstream processing toward task-appropriate responses [7]. Temporal sites (TP9, TP10) additionally contribute through audiovisual integration and semantic processing of incoming content. Within the present paradigm, these four electrodes therefore cover the two cortical regions most theoretically implicated in the moment-to-moment appraisal of whether a video warrants continued attention. *Search Query:* `"frontal lobe" OR "temporal lobe" "EEG" "rapid decision making" OR "visual appraisal"`

2.3 Pre-Motor Intent and Readiness Potentials

- **Claim:** The cognitive decision to voluntarily act (e.g., a thumb swipe) forms in the cortex significantly before the physical motor action is executed, known as the Readiness Potential (Bereitschaftspotential) [8,9].

- **Claim:** Isolating the 1–3 seconds preceding a mechanical action effectively captures the core cognitive decision while decoupling it from the downstream physical motor artifacts of the arm movement.

Search Query: `"pre-motor" EEG "decision making" "motor artifact" isolation`

- **Claim:** A physical keystroke or swipe is merely the delayed behavioral exhaust of an engagement failure that has already occurred neurologically.
Search Query: `"motor execution delay" "cognitive decision" intention EEG`

2.4 Hardware Validity and Experimental Design

- **Claim:** While neuroimaging research has demonstrated that neural signals measured during video viewing predict individual engagement decisions and aggregate viewing behavior above and beyond self-report and behavioral measures [10], the controlled laboratory constraints inherent to fMRI preclude naturalistic paradigms; the present study therefore deliberately employs consumer-grade EEG to enable the ecological validity the research question demands, at the cost of spatial resolution and subcortical access.

- **Claim:** Consumer-grade, 4-channel dry-electrode EEG headsets (like the Muse S) provide valid, reliable, research-grade spectral data for cognitive state classification. Krigolson et al. demonstrated that the MUSE system yields statistically quantifiable ERP components (N200, P300, reward positivity) at electrodes AF7, AF8, TP9, and TP10 that are indistinguishable from those recorded with a research-grade 64-channel ActiChamp system [11]. This validation is bounded to low-frequency ERP components (<15 Hz); the present study's empirical ablation (Step 3) provides the direct evidence that the device captures meaningful variance in the high-gamma range under naturalistic conditions.
Search Query: `"Muse EEG" OR "consumer-grade EEG" validity "cognitive state classification"` The Muse device has been adopted in peer-reviewed EEG-based user engagement research [12]. In a controlled learning task using a dry-electrode wearable system, Apicella et al. achieved 76.9% within-subject accuracy for cognitive engagement classification via a Filter Bank–CSP–SVM pipeline, establishing that few dry electrodes are sufficient for engagement detection in a structured paradigm [13]. The present study extends this precedent to an unstruc-

tured naturalistic browsing context, where stimulus control is deliberately surrendered in favor of ecological validity. Within the same institutional research group, Moontaha et al. developed a real-time emotion classification pipeline using the identical Muse S hardware, achieving mean F1-scores of 87% and 82% for arousal and valence recognition respectively under immediate label conditions from consumer-grade EEG with controlled video stimuli [14]. The present study shares the hardware platform and spectral power feature extraction approach, but diverges fundamentally in objective—predicting binary engagement micro-decisions rather than continuous valence-arousal affect dimensions—and in paradigm design, replacing controlled emotional video elicitation with unconstrained algorithmic feed browsing.

- **Claim:** Naturalistic, unconstrained "free-browsing" experimental designs provide critical ecological validity that cannot be replicated in strictly constrained, artificial lab environments. Hasson et al. established the scientific legitimacy of this trade-off by demonstrating that subjects freely viewing an unedited feature film—with no fixation constraint, no trial structure, and no controlled stimulus onset—produced extensive, highly significant cortical synchronization across individuals, concluding that "natural vision drastically differs from conventional visual mapping studies" by relaxing constraints on stimulus isolation, object motion, eye movements, and multimodal interaction [15]. Critically, the same study identified that higher-order association areas—including the supramarginal gyrus, angular gyrus, and prefrontal cortex—consistently *failed* to show intersubject coherence, with the authors explicitly attributing this dissociation to those regions being "linked to unique, individual variations" rather than stereotyped, stimulus-driven responses [15].
  Search Query: `"ecological validity" "naturalistic" EEG experiment "free-viewing" OR browsing` Bellos et al. identify audiovisual and media consumption as established EEG-based user engagement research domains, confirming the scientific legitimacy of stimulus-driven naturalistic paradigms [12].

- **Claim:** Modern neuroergonomics recognizes that surrendering strict experimental stimulus control (e.g., allowing algorithmic feed curation) is a necessary and mathematically acceptable trade-off to capture authentic cognitive states.
  Search Query: `"neuroergonomics" "ecological validity" "stimulus control" trade-off naturalistic EEG`

2.5 Signal Processing and High-Frequency Legitimacy

- **Claim:** Baseline relative power normalization effectively cancels out idiosyncratic physiological noise such as skull thickness and basal resting voltage differences across subjects.
  Search Query: `"EEG" "relative power" OR "baseline normalization" "individual differences"`

- **Claim:** High Gamma (40–60 Hz) activity carries genuine, vital neurological variance related to complex cognitive binding, rather than just being muscular or electrical noise. Bellos et al. establish in their systematic review that gamma waves (30–100 Hz) are associated with high levels of cognitive processing, perception, and attention, and with peak emotional experiences [12]. At the systems neuroscience level, gamma-band synchronization has been proposed as the fundamental mechanism through which attentional selection is implemented in cortical circuits, with attended stimuli producing stronger and higher-frequency gamma responses [16]. At the hardware level, Pattisapu and Ray demonstrated that a low-cost EEG amplifier (OpenBCI) can detect stimulus-induced narrow-band gamma oscillations (30–70 Hz) with spectral and temporal profiles significantly correlated with those of a research-grade Brain Products system (Spearman $\rho = 0.94$ for slow gamma, $\rho = 0.75$ for fast gamma) [17]. The present ablation study extends this principle to the dry-electrode naturalistic context: if the 40–60 Hz band contained only power-line artifact, classification performance would collapse upon notch filter removal; the marginal improvement observed instead constitutes direct empirical evidence that the Muse S captures recoverable cognitive variance in this range.
  Search Query: `"High Gamma" 40-60 Hz EEG "cognitive binding" OR "decision making"`

- **Claim:** The standard 50 Hz power-line notch filter can be overly destructive, ablating vital high-frequency cognitive data if misapplied in BCI contexts.
  Search Query: `"50 Hz notch filter" "EEG" "gamma band" data loss`

- **Claim:** Distilling volatile time-series arrays into geometric/statistical variance parameters (mean, std, min, max) acts as a powerful regularizer for EEG classification against temporal jitter and phase noise.
  Search Query: `"EEG feature extraction" "statistical features" OR "variance" time-series phase noise`

2.6 Machine Learning, Idiosyncrasy, and BCI Generalizability

- **Claim:** High-dimensional non-linear ensembles (like Random Forests) significantly outperform simple linear models when decoding dynamic, multi-channel EEG cognitive states.
Search Query: `"Random Forest" EEG "cognitive state" classification "high dimensional"`

- **Claim:** In small-sample EEG classification, deep learning architectures (like Transformers) often fail to outperform carefully feature-engineered classical ensembles (like Random Forests) due to their susceptibility to overfitting on high-dimensional noise.
Search Query: `"EEG classification" "Random Forest" vs "deep learning" OR Transformer small sample size`

- **Claim:** Even when controlling for absolute physiological voltage differences via baseline normalization, cross-subject BCI generalizability (Leave-One-Subject-Out) fails because the functional cortical strategies encoding engagement are fundamentally unique to the individual.
Search Query: `"EEG" BCI "cross-subject generalizability" OR "Leave-One-Subject-Out" "baseline normalization" idiosyncratic`

- **Claim:** Universal, static equations (like the SFEI) are mathematically insufficient for mapping complex short-form media engagement compared to individualized machine learning calibration.
Search Query: `"EEG" BCI "individualized calibration" vs "universal models" OR static`

- **Claim:** Rigorous handling of class imbalance (e.g., 50/50 prior distribution enforcement) is critical to prevent base-rate classifier bias in overlapping sliding-window EEG data.
Search Query: `"EEG classification" "class imbalance" "sliding window" undersampling`

2.7 Behavioral Sequence Heterogeneity and Attentional Modes

- **Claim:** User behavior in continuous media consumption exhibits temporal autocorrelation, where the probability of skipping is sequentially dependent on prior actions rather than being purely isolated content

appraisals.
Search Query: `"sequential dependence" OR autocorrelation "media consumption" OR browsing behavior`

- **Claim:** Cognitive disengagement during continuous tasks is highly heterogeneous, comprising both active, stimulus-locked rejection and a generalized, state-driven "browsing" momentum. Anderson and Wood established that habitual social media users exhibit reward-insensitive posting behavior controlled by contextual cues rather than content appraisal [18]. Knierim et al. further demonstrated that passive feed scrolling produces a behaviorally distinct pseudo-flow state—absorption without challenge or goal-directedness—that is phenomenologically different from genuine content-locked engagement [3]. Baughan et al. provide the behavioral phenomenology underlying this heterogeneity: some skip events correspond to active, content-locked rejection (users who disliked a specific post), while others reflect a state of normative dissociation in which the user was scrolling on autopilot with no conscious content appraisal occurring at all [2]. These two cognitive states are neurologically distinct, which directly predicts the class heterogeneity observed in the present data: the `SKIP` class captures both genuine content-locked appraisal failures and a generalized disengaged browsing momentum in which the decision to skip is state-driven rather than stimulus-locked. *Search Query:* `"cognitive disengagement" heterogeneity "attentional state" continuous performance task`

- **Claim:** There is a logical error possibly: firstly we need to be clear about what our two classes cover cause i think that there is a mismapping to what they mean. also, posting? why posting? thats weird and out of context. and also, liking? may be relevant, but as you know we explicitly told people not to interact (eg. like) with infinite tiktok feed and just do the swiping and clicking button at the same time.

Chapter 3

Materials and Methods

3.1 Participants

Twenty-eight ($N = 28$) participants were recruited for the study ([Insert N_F female, N_M male]). To participate, individuals were required to have normal or corrected-to-normal vision and no history of neurological disorders. Three participants were excluded *a priori*: two due to a high number of self-reported keypress errors (greater than 10) which compromised label integrity, and one due to an undisclosed neurological condition. The experimenter's own data, although collected for pipeline validation, was strictly excluded from all reported results to eliminate experimenter bias. The final sample consisted of $n = 25$ participants (age range 20–30 years, $M = 24.6$, $SD = 2.9$).

Participants were collected over a total of five days and were conceptually divided into two groups based on recruitment timing: an initial smaller group was recorded in a separate room without financial compensation, while the subsequent main group (recorded a few days later) received a 20-euro voucher as an incentive. All participants provided informed consent and signed data protection agreements approved by the Ethics Committee of the University of Potsdam prior to the study (INSERT: ethics proposal number).

[Insert tex table of participants, answers and co]

3.2 Data Acquisition

Electroencephalography (EEG) data were recorded using a consumer-grade Muse S headband (Generation 2) Figure 3.1. The device features four dry electrodes placed at positions AF7, AF8, TP9, and TP10 according to the international 10–20 system, with the reference electrode located at Fpz. Data were continuously sampled at an average rate of 256 Hz.

[Insert Picture of Muse Headband]

Figure 3.1: The consumer-grade Muse S (Gen 2) EEG headband used for data acquisition.

The EEG signals were transmitted via Bluetooth and recorded using the Lab Streaming Layer (LSL) protocol with the `muselsl` library. A custom Python recording script was developed to handle data collection (INSERT: reference to exact github code link of script). This script featured real-time frequency monitoring and an automatic reconnection mechanism to seamlessly manage any Bluetooth drops without disrupting the session.

To synchronize the behavioral data (video transitions) with the continuous EEG stream without relying on video monitoring of the participant, keystroke markers were mapped directly into the recording script. Participants used their non-dominant hand to press a specific key on a laptop keyboard provided in the room: the "B" key to denote the start of the baseline period, and the "A" key to mark every manual swipe (skip event) during the TikTok browsing session.

[Insert Picture of Experimental Setup Cabin]

Figure 3.2: The experimental setup cabin ensuring privacy and a distraction-free environment for the participant.

[Insert Closer Picture of Laptop and Phone Setup]

Figure 3.3: Close-up of the recording setup showing the researcher's phone and the laptop used for keypress synchronization.

3.3 Experimental Protocol

The experimental sessions were scheduled for 45-minute timeslots, with the actual experimental procedures taking approximately 35 to 40 minutes. Upon arrival, participants were greeted, provided with an information sheet, and asked to sign the ethics consent forms. Afterward, they completed a pre-experiment demographic and basic screening survey, which took approximately five to ten minutes.

Participants were then instructed on the tasks. The Muse S headband was mounted on their head and adjusted to ensure optimal skin contact, particularly on the forehead and behind the ears for participants with long

[Insert Schematic Flowchart of Experimental Protocol]

Figure 3.4: Experimental protocol flow: Introduction and consent, EEG setup, 2-minute baseline video, task explanation, and 25-minute TikTok free browsing session.

hair. The experimenter verified the signal quality and ensured the device was adequately charged.

The experimental paradigm consisted of two main phases: a baseline recording and a free-browsing session. For the baseline phase, participants were provided with the researcher's iPhone, and a two-minute nature video on YouTube was started. Participants were instructed to relax and simply watch the video. Concurrently, the recording script was initiated, and the "B" marker was pressed. To ensure ecological validity while keeping the participant focused, the iPhone's "Guided Access" feature was enabled to prevent exiting the application. Participants were left alone in the room to afford them privacy. After two minutes, the Guided Access lock visually timed out, signaling the conclusion of the baseline period.

The experimenter then returned to explain the second phase: a naturalistic short-form video browsing task using the TikTok application. To ensure a standardized yet personalized starting state, the app's "For You" feed was refreshed using the "Reset feed" option in the content preferences settings. Participants were instructed to consume TikTok content as they normally would, watching videos they found interesting and swiping away those they found boring. To restrict the action space strictly to scrolling behavior, they were asked to refrain from using the "like" function, writing comments, or visiting user profiles.

Crucially, participants were tasked with pressing the "A" button simultaneously with every swipe motion on the phone to accurately synchronize their micro-decisions with the EEG data. For instance, a participant holding the phone in their right hand used their left hand to press the designated key on the adjacent laptop keyboard.

[Insert Schematic of Button Click Synchronization]

Figure 3.5: Schematic representation of the manual synchronization process: the participant swipes to skip the video and simultaneously presses the 'A' button to mark the event in the EEG stream.

The Guided Access feature was reactivated and set to a 25-minute limit for the TikTok app. Participants were reminded that the headband script would automatically reconnect if a Bluetooth disconnection occurred. They were then left alone in the room with the door closed, creating an isolated viewing environment designed to elicit natural responses to the video feed. They were instructed to knock on the glass door if any errors or issues arose; however, no significant interruptions occurred during the sessions.

Once the 25-minute viewing task concluded, the experimenter entered the room, and the participant removed the headband. They then answered a few post-session survey questions, specifically self-reporting if they made any keypress synchronization errors during the session. Finally, they signed the required compensation forms to receive their 20-euro voucher. The post-measurement debrief and surveying process took roughly two to four minutes. Participants were thanked for their time, briefly debriefed about the broader context of the study if interested, and dismissed.

3.4 Labeling Strategy

To quantify class separability under different labeling choices, Cohen's d was computed as the standardized mean difference between the STAY and SKIP distributions. A parameter sensitivity analysis was then performed in which gap length, window length, and imminent-skip period were varied systematically. When gap length and window length were held constant, increasing the imminent-skip period produced a corresponding increase in Cohen's d, indicating progressively stronger class separation.

3.4.1 Operational Definition of Labeling Parameters

The labeling strategy was derived directly from the research questions and the aim of isolating neural states relevant to sustained engagement and imminent disengagement during naturalistic browsing. Accordingly, the computational target was constrained to two discrete classes: sustained engagement (`STAY`) and impending disengagement (`SKIP`).

The default labeling scheme was defined by three temporal parameters. The gap length was set to 2.0 seconds, the window length was set to 1.0 second, and the imminent-skip period was set to 3.0 seconds. In the default configuration, the `SKIP` class therefore comprised the 1.0-second window immediately preceding the 2.0-second gap before each registered `A`-keypress (swipe event), yielding a label placement that aligned with the pre-skip attentional state while avoiding the most immediate motor-preparatory interval. The `STAY` class comprised all remaining continuous viewing segments that satisfied the exclusion criteria described below.

3.4.2 Operational Definition of STAY and SKIP

The `SKIP` class was defined as the finite 3.0-second window terminating 2 seconds before the timestamp of each registered `A`-keypress (swipe event). This window captures neural activity preceding the physical act of skipping a video while excluding the motor-preparation interval, which has been reported to occur approximately 500–1500 ms before movement onset; this interval is therefore covered by the 2-second buffer preceding the swipe time marker.

The `STAY` class represents sustained cognitive engagement. It was defined by extracting all remaining continuous viewing periods during the naturalistic TikTok consumption phases that were i) outside the 3.0-second `SKIP` window, ii) outside the 2-second pre-skip motor-preparation buffer, and iii) outside the 0.5-second post-skip orientation grace period.

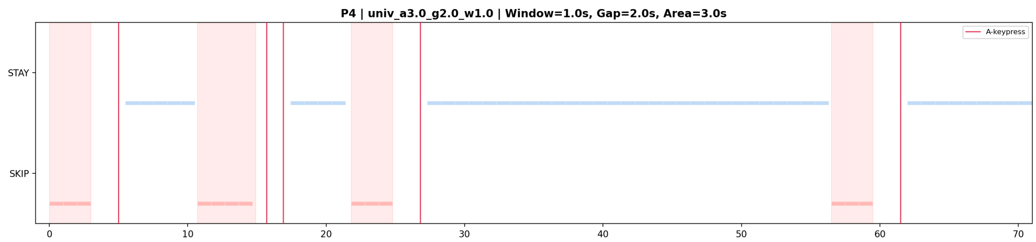

Figure 3.6: Schematic overview of the labeling strategy under the default parameterization. The figure illustrates the placement of the `STAY` and `SKIP` labels relative to the registered press markers, including the default gap length, window length, and imminent-skip period.

3.4.3 Baseline Exclusion

The manually categorized pre-task resting baseline phases were excluded from the `STAY`/`SKIP` classification dataset and used only for per-participant signal normalization. Consequently, the algorithms were trained to discriminate between active engagement (`STAY`) and approaching disengagement (`SKIP`) during browsing, rather than between resting and viewing. Figure 3.6 illustrates the default labeling logic, including the press markers and the temporal placement of the labeling windows.

In addition to baseline exclusion, the labeling procedure omitted segments affected by boundary overlap between `STAY` and `SKIP` definitions. This was necessary to preserve clear class assignment and avoid ambiguous temporal regions near each swipe event. The final label construction therefore prioritized temporal purity of the target classes over maximal sample count.

3.4.4 Parameter Sensitivity Analysis

To assess how label definition affects class separability, a controlled sensitivity analysis over three temporal parameters was conducted. In each experiment, two parameters were held constant while the third was varied, isolating its effect. This setup allows us to distinguish whether separability is primarily driven by temporal distance to the swipe event, window size, or their interaction.

The influence of window size (ws), gap size (gs), and imminent-skip period (isp) on data preprocessing and segmentation is illustrated in three example-configurations: (i) default parameters (ws = 1s, gs = 2s, isp = 3s; Figure 3.7), (ii) reduced window size (ws = 0.3s, gs = 2s, isp = 3s; Figure 3.8), and (iii) reduced gap with extended imminent-skip period (ws = 1s, gs = 0.5s, isp = 4s; Figure 3.9). The swipe event is indicated by vertical red lines. All figures show the same randomly selected 70-second segment from a single participant. Time (seconds) is plotted on the x-axis, and class labels—STAY (blue) and SKIP (red) on the y-axis.

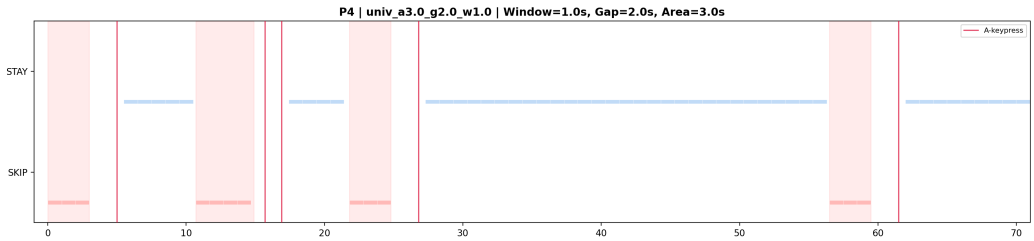

Figure 3.7: Window parameter exploration (default parameters, same as Figure 3.6).

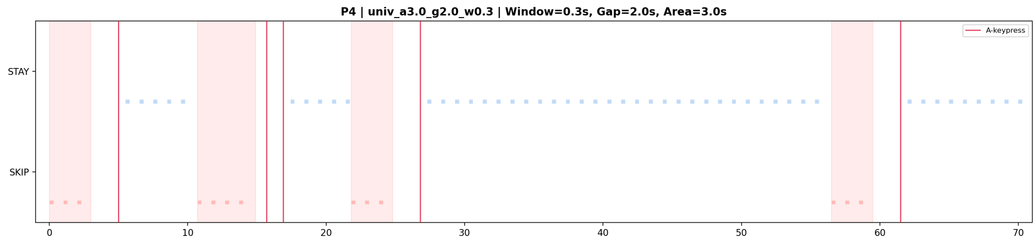

Figure 3.8: Window parameter exploration (default parameters, except a smaller window size).

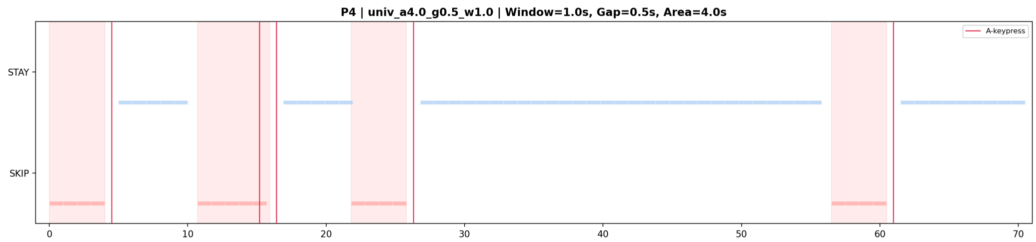

Figure 3.9: Window parameter exploration (default parameters, except a smaller gap size & a longer imminent-skip period (Skip-Area)).

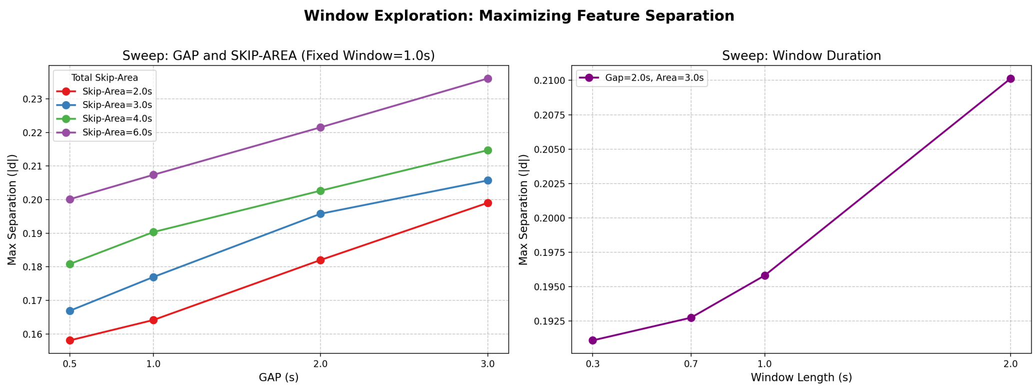

Figure 3.10: Parameter sweep of the window exploration analysis.

The key result is that, with fixed imminent-skip period and window size, increasing the gap length consistently increases Cohen's d (Figure 3.10, left), indicating greater separability as the SKIP interval moves further from the swipe event. A similar effect is observed for longer imminent-skip periods, where larger SKIP intervals also improve separability. For fixed gap and imminent-skip period, increasing window size shows a roughly linear or potentially cubic relationship with Cohen's d (Figure 3.10, right), again corresponding to increased class separation. While all these results strongly suggest better separability with increased overall data-availability (bigger windows, and areas) and simultaneously better separability the further we move away from the swipe, interpreting these relationships and their implications for optimal parameter selection remains beyond the scope of this thesis as it requires separate analysis, and is therefore a promising direction for future work. To maintain focus and scope, this analysis proceeds with the locked, default parameters:

- Window size (ws) = 1 second
- Gap size (gs) = 2 seconds
- Imminent-skip period (isp) = 3 seconds

3.4.5 Class Separation Metric

Cohen's d was used as the primary descriptive measure of class separation. It expresses the standardized difference between the STAY and SKIP distributions, making it suitable for comparing separability across parameter settings even when the underlying distributions differ in scale. In this context, larger absolute values of Cohen's d indicate stronger separation between the two classes.

$$d = \frac{\bar{x}_{\text{STAY}} - \bar{x}_{\text{SKIP}}}{\sqrt{\frac{(n_{\text{STAY}}-1)s^2_{\text{STAY}}+(n_{\text{SKIP}}-1)s^2_{\text{SKIP}}}{n_{\text{STAY}}+n_{\text{SKIP}}-2}}} \tag{3.1}$$

where $\bar{x}_{\text{STAY}}$ and $\bar{x}_{\text{SKIP}}$ denote the sample means of the feature values for the STAY and SKIP windows respectively, s^2_{STAY} and s^2_{SKIP} are the corresponding sample variances, and n_{STAY} and n_{SKIP} represent the number of samples in each class.

For each parameter configuration, Cohen's d was computed between the STAY and SKIP feature distributions derived from the corresponding labeling scheme. This provided a direct quantitative index of how strongly the chosen temporal definitions emphasized the contrast between sustained engagement and imminent disengagement.

3.5 Universal Recording Pipeline

To facilitate rigorous and reproducible data collection, a robust Python-based recording architecture (`scripts/recording_script_v4.py`) was developed. This script orchestrates the simultaneous capture of raw EEG telemetry from the Muse S headband and has the capability of recording synchronized behavioral video from a webcam which was not used in this experiment, yet may be of value to future investigations. Its autonomous capabilities ensure that the experimental protocol can be executed reliably without extensive technical oversight.

For this study's recordings the command `pythonrecording_script_v4.py--nocamera--duration1800` was used to call the script. The two flags simply mean (i) bypassing camera recording, as it is not in the scope of this study yet kept for future investigations and (ii) the recording duration is 1800s = 30 minutes. The pipeline's core features include:

- **Automated Hardware Discovery:** Upon launch, the script automatically scans the local Bluetooth environment for active Muse devices and enumerates all connected camera peripherals, allowing the researcher to select the optimal hardware interactively without requiring hardcoded MAC addresses or device indices.

- **Precision Synchronization and Event Logging:** A dedicated, threaded keypress listener (`pynput`) monitors the `A` and `B` keys used by participants to mark swipe events and baseline respectively. Crucially, the script computes a dynamic time offset between the system clock and the LSL stream clock. Keypresses are retroactively aligned to this LSL timeline within a defined tolerance window and embedded directly into the EEG data rows as binary flags, ensuring absolute temporal synchrony between cognitive events and physical interactions.

- **Simultaneous Multimodal Capture:** The script utilizes the `pylsl` library to interface with the Lab Streaming Layer (LSL), pulling raw EEG data from the Muse headband at 256 Hz. Unless deactivated with the flag, it concurrently employs OpenCV to capture and encode synchronized webcam video using the H.264 (avc1) codec, providing visual ground truth of participant behavior.

- **Resilient Auto-Reconnection:** To combat the inherently unstable nature of Bluetooth Low Energy (BLE) connections, the script features an active connection-loss detection loop. If the LSL stream halts (no samples received within a timeout period), the script automatically terminates the hanging subprocess, rescans the Bluetooth environment, restarts the stream, and initiates a sequential data fragment (e.g., `_2.csv`, `_3.csv`). This guarantees that transient connection drops do not corrupt the entire recording session.

- **Concurrent Multi-Threading:** To ensure that blocking operations (such as writing high-resolution video frames to disk) do not disrupt the high-frequency polling required for EEG telemetry, the script isolates the LSL pulling loop, the OpenCV capture loop, and the keypress listener into independent Python daemon threads.

This open-source compatible architecture ensures that any independent researcher equipped with a standard webcam and a consumer-grade Muse S headband can fully replicate the data collection methodology utilized in this thesis.

3.6 Universal Pre-Processing Pipeline

The following subsections describe step-by-step exactly what the Python code in the `05-12_data_analysis_and_results/scripts` directory executes to process the raw EEG signals into a uniform shape suitable for analysis.

3.6.1 Data Synchronization and Interpolation

The preprocessing pipeline began with the chronological alignment and concatenation of all recorded CSV files for each participant, ensuring a continuous temporal sequence (executed via `05-12_data_analysis_and_results/scripts/step01_preprocess.py`). The raw data streams collected from the LSL (Lab Streaming Layer) exhibited slight variations in intrinsic sampling rates. To establish a uniform temporal grid suitable for across-session and across-participant analysis, the true sampling rate was derived by calculating the median time difference (Δt) between consecutive timestamps. Subsequently, a linear interpolation algorithm was applied to upsample and regularize the data to a constant sampling frequency of exactly $f_s = 256$ Hz. This process guarantees that any subsequent spectral analysis relies on a strictly uniform time axis, avoiding mathematical artifacts stemming from temporal jitter.

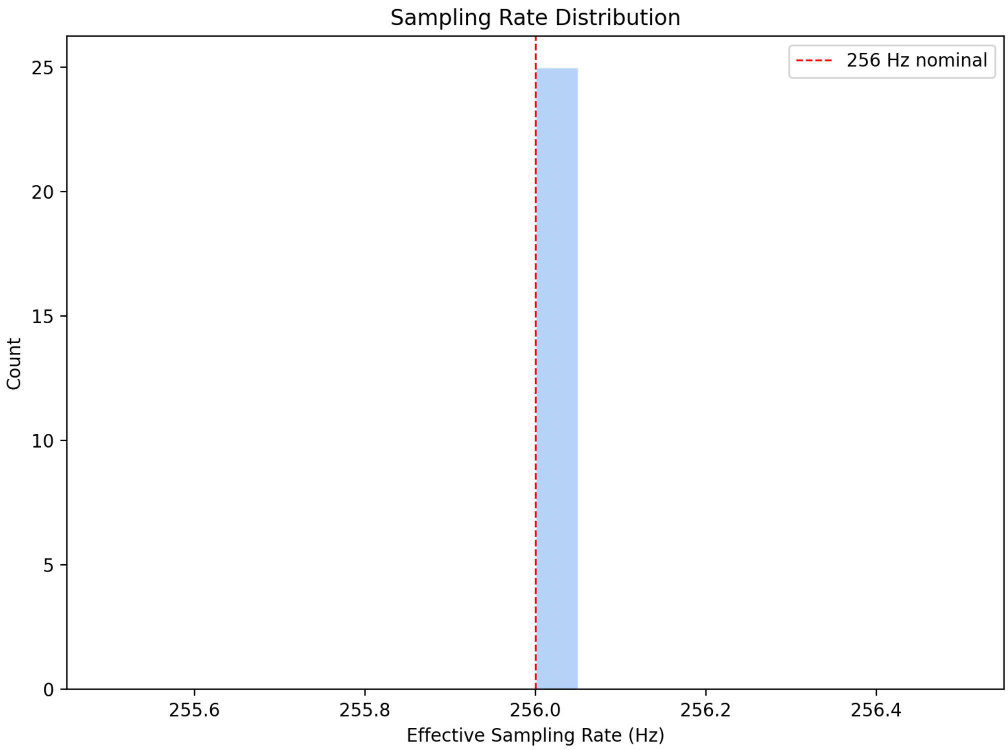

Figure 3.11: Sampling Rate Distribution across participants, demonstrating that after interpolation all signals operate precisely at 256 Hz, which is a mathematical prerequisite for the correct extraction of spectral band power.

During this temporal alignment process, Bluetooth transmission dropouts were identified by locating unnatural temporal gaps between consecutive CSV boundaries. To ensure data integrity, the duration of these dropouts was logged, and any potential data windows traversing such boundaries were strictly excluded in later extraction phases.

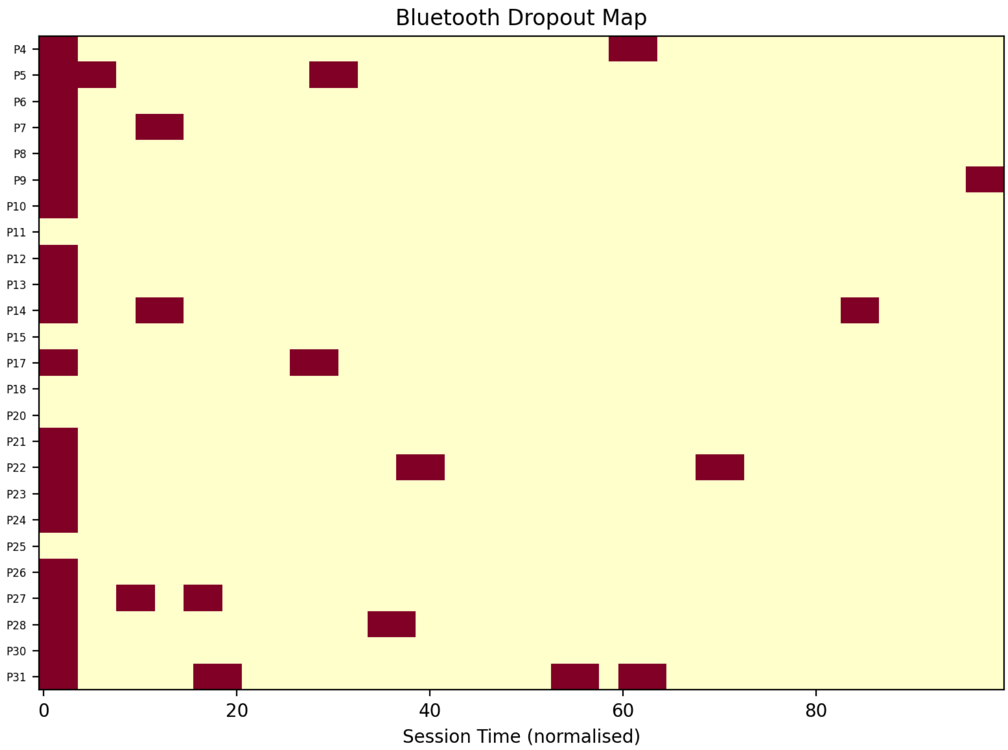

Figure 3.12: Bluetooth Dropout Heatmap indicating signal loss events across the experimental timeline. Data windows overlapping with these dropouts were removed from the analysis pipeline.

3.6.2 Baseline Extraction and Normalization

Following temporal synchronization, a baseline resting state was extracted for each participant to account for individual physiological differences, such as varying skull thickness or inherent resting cortical activity. The baseline period was rigidly defined as a 90-second continuous window commencing 10 seconds after the participant's initial `B` keypress (executed via `05-12_data_analysis_and_results/scripts/step01_preprocess.py`).

The interpolated signal was then separated into seven standard EEG frequency bands (Delta, Theta, Alpha, Beta, Low Gamma, High Gamma, and Very High Gamma) utilizing a 4th-order Butterworth bandpass filter. The instantaneous amplitude envelope for each band was extracted and squared to compute the instantaneous spectral power P.

To normalize the extracted features across participants, a z-score relative power normalization was applied against the participant-specific baseline. For every temporal sample, the normalized spectral power z was computed as:

$$z = \frac{P - \mu_{base}}{\sigma_{base}} \tag{3.2}$$

where μ_{base} and σ_{base} denote the mean and standard deviation of the spectral power derived exclusively from the aforementioned 90-second baseline period. This relative power transformation ensures that all subsequent statistical evaluations and machine learning models operate on normalized deviations from a participant's own resting state, effectively mitigating absolute inter-subject voltage variance.

3.6.3 Windowing and Class Labeling

Following normalization, the continuous data streams were segmented and labeled into discrete `STAY` and `SKIP` classes based on the participant's interaction events (`A` keypresses) as previously established in Sections 3.4.1 and 3.4.2. To strictly prevent motor artifact contamination arising from the physical swipe action, a **2.0-second gap** was enforced immediately preceding the `A` keypress, and a **0.5-second gap** was enforced immediately following it. No data was extracted from these gap periods.

The `SKIP` class was rigorously defined as the **3.0-second window** immediately preceding the aforementioned 2.0-second pre-swipe gap (executed via `05-12_data_analysis_and_results/scripts/step03_label_windows.py`). This temporal framing specifically isolates the critical cognitive decision-making phase leading up to the swipe, entirely uncontaminated by the physical execution of the swipe itself. The `STAY` class was defined as all remaining valid temporal periods outside of the `SKIP` windows and safety gaps.

Finally, discrete 1.0-second analysis windows were extracted using a sliding window with no overlap (stride = window length). This ensures that windows do not cross class boundaries, enter safety gaps, or overlap with adjacent samples, thereby preventing data leakage and artificially inflated model performance.

3.6.4 Artifact Rejection

To ensure the integrity of the analyzed EEG signals, an automated outlier detection approach was implemented to flag potential artifacts (executed via `05-12_data_analysis_and_results/scripts/step02_artifact_flag.py`). Specifically, the 95th percentile of the peak-to-peak amplitude computed on the notch-filtered signal was utilized as a dynamic, participant-specific threshold. Frontal channels (AF7, AF8) were evaluated for ocular artifacts (e.g., blinks), while temporal channels (TP9, TP10) were evaluated for electromyogenic (EMG) artifacts. Absolute voltage thresholds (e.g., $150\mu V$) were initially investigated but were ultimately discarded as they did not yield generalizable or useful results across the dataset.

No data points were excluded from the final machine learning pipeline based on these artifact flags. Incorporating such filtering would introduce additional experimental conditions and substantially expand the analysis beyond the scope of this proof-of-concept study. Instead, the approach is documented to highlight awareness of artifact-related confounds and to provide a foundation for more advanced rejection strategies in future work.

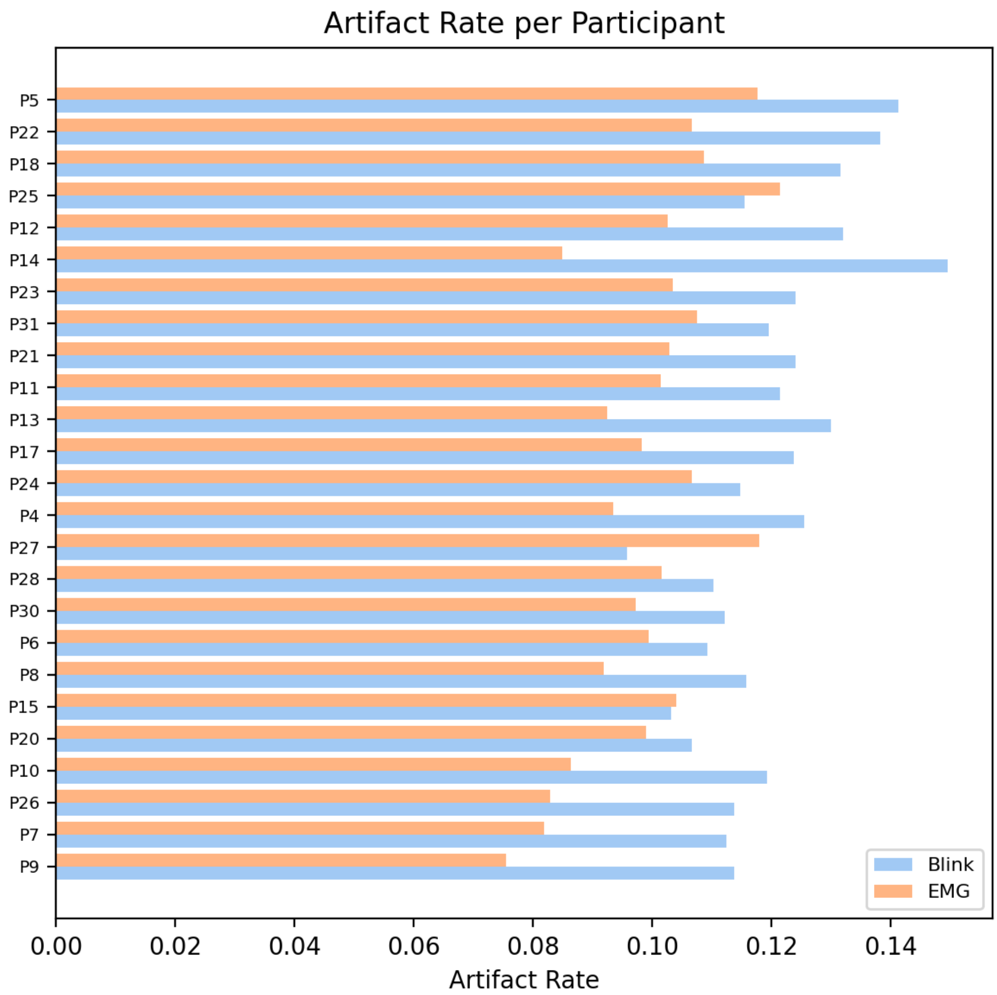

Figure 3.13: Artifact rate distribution and signal quality flagging based on the 95th percentile peak-to-peak outlier approach.

3.6.5 Behavioral Burst Rejection

In addition to physiological artifacts, behavioral anomalies termed "bursts" were identified. A burst is defined as a sequence of rapidly successive skip events (e.g., multiple `A` keypresses merging into a single contiguous temporal block), reflecting frantic swiping rather than deliberate evaluation. Flagging these events aims to ensure that the `SKIP` class captures genuine cognitive disengagement rather than high-momentum, low-engagement scrolling behavior.

Although bursts were successfully detected, they were not excluded from the final analysis. Figure 3.14 visualizes the detection mechanism via the distribution of inter-swipe intervals (i.e., dwell time) across all participants. The vertical threshold marks the 6.0-second merge boundary ($2\times$the 3.0-second `SKIP` window). Consecutive swipes below this threshold cause their respective `SKIP` windows to overlap and merge into a single contiguous block, thereby defining a burst. The shaded region highlights the frequency of such rapid, consecutive events.

It is important to note that the `SKIP` window size is defined independently of the dwell time distribution; the 3.0-second value is an intuitive choice. As such, its influence on the resulting burst structure warrants further investigation. Incorporating burst rejection would introduce additional analytical complexity and is therefore beyond the scope of this proof-of-concept study. Instead, its inclusion highlights a relevant behavioral confound and motivates future work.

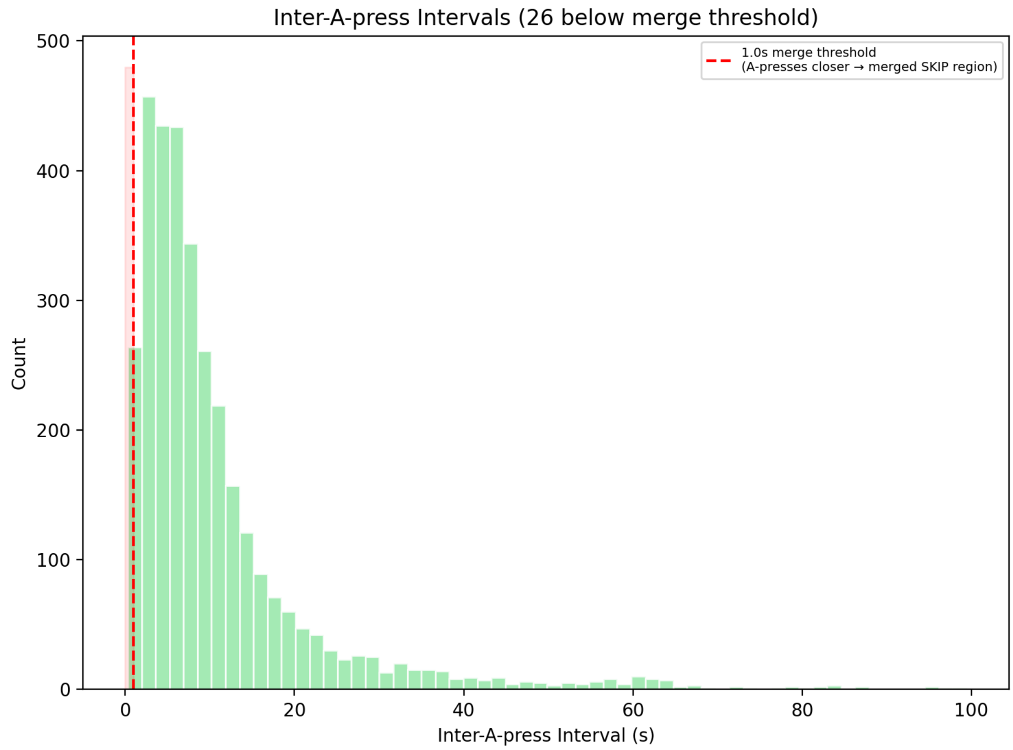

Figure 3.14: Distribution of inter-swipe intervals (view times of each video across all participants, or dwell time). The shaded region highlights intervals below the 6.0-second merge threshold, where consecutive `SKIP` windows overlap to form behavioral burst blocks.

3.7 Feature Engineering

3.7.1 Band Power and Statistical Moments

Following window extraction, a comprehensive feature set was computed for each valid temporal window (executed via `05-12_data_analysis_and_results/scripts/step05_feature_engineering.py`). For each of the 4 electrodes across the 7 previously defined frequency bands, four primary statistical moments were extracted: Mean, Standard Deviation, Minimum, and Maximum. An example of these extracted statistical moments belonging to one temporal 1-second window of one frequency band (alpha) of one electrode of one participant, is depicted in Figure 3.15.

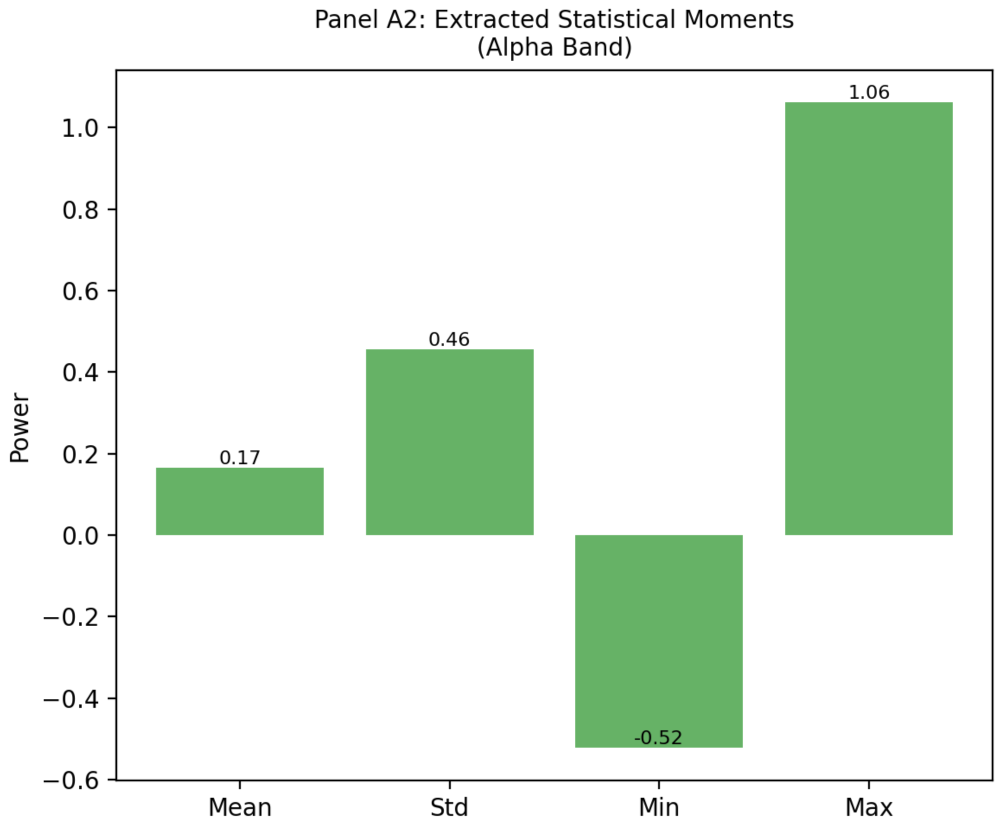

Figure 3.15: Extraction of standard statistical moments from the normalized EEG frequency bands.

3.7.2 Engagement Index Calculation

To provide a recognized clinical baseline for cognitive evaluation, the Engagement Index (EI) was calculated for each analysis window. The EI is theoretically grounded in established neurophysiological research correlating task engagement with frontal and parietal band power ratios. It is mathematically defined as:

$$EI = \frac{\beta}{\alpha + \theta} \tag{3.3}$$

In this pipeline, exactly one scalar Engagement Index (EI) is computed per 1.0-second analysis window per participant. To calculate this single value, the signal variance (representing total band power) is first extracted for the Beta (β), Alpha (α), and Theta (θ) bands across each of the four individual electrodes. These four spatial variances are then averaged together to produce a single, whole-head β, α, and θ scalar for that specific window, which are finally fed into the EI ratio formula. This single composite scalar serves as the primary clinical benchmark against which the predictive power of individual statistical features is evaluated in the machine learning phase. An example calculation of the EI for a single sample window of one participant can be observed in Figure 3.16, visually showing the three spatially-averaged scalars that are utilized in the Index calculation in Formula 3.3.

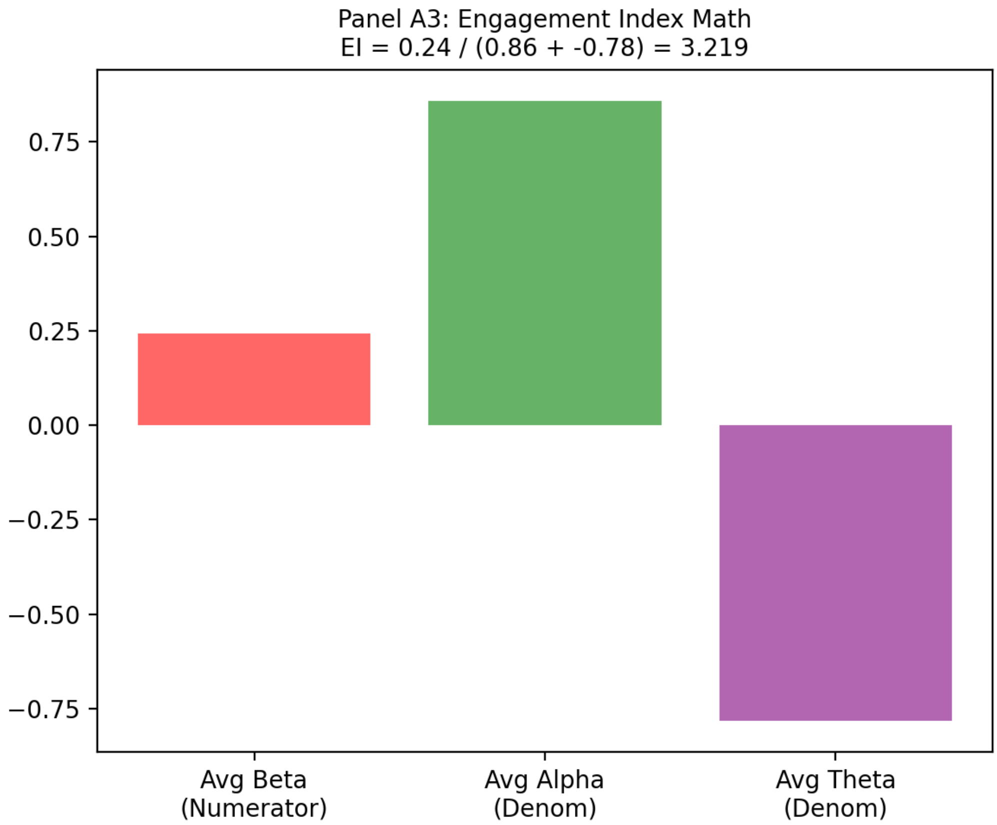

Figure 3.16: Engagement Index (EI) calculation deriving a single scalar engagement metric from the Beta, Alpha, and Theta band powers.

3.7.3 Hjorth Parameters

In addition to basic descriptive statistics, Hjorth Parameters (Activity, Mobility, and Complexity) were computed to capture the sequence-dependent temporal dynamics of the signal within the 1.0-second window. While standard statistical moments characterize the amplitude distribution independently of their temporal order (i.e., they are time-invariant), Hjorth parameters depend explicitly on the time-domain derivatives, thus capturing spectral properties—such as mean frequency and bandwidth—directly from the time-domain signal.

For a discrete EEG signal vector $y(t)$ of length $N = 256$ representing a specific channel and frequency band within a window, the three parameters are mathematically defined as follows:

Activity

Activity measures the total signal power and is defined simply as the variance of the signal amplitude:

$$\text{Activity} = \sigma_y^2 = \frac{1}{N}\sum_{t=1}^{N}(y(t) - \mu_y)^2 \tag{3.4}$$

where μ_y is the mean of $y(t)$. The input dimension is the normalized amplitude and the output dimension represents the squared amplitude.

Mobility

Mobility serves as a time-domain proxy for the mean frequency of the power spectrum. It is defined as the square root of the ratio of the variance of the first derivative $y'(t)$ to the variance of the primary signal $y(t)$:

$$\text{Mobility} = \frac{\sigma_{y'}}{\sigma_y} = \sqrt{\frac{\text{var}(y'(t))}{\text{var}(y(t))}} \tag{3.5}$$

Since the differentiation of a signal with respect to time yields a rate of change, the resulting output dimension of Mobility is expressed in Hertz (Hz).

Complexity

Complexity measures the bandwidth of the signal and indicates how closely the signal's shape resembles a pure sine wave (where a pure sine wave converges to a complexity of 1). It is defined as the ratio of the Mobility of the first derivative $y'(t)$ to the Mobility of the primary signal $y(t)$:

$$\text{Complexity} = \frac{\text{Mobility}(y')}{\text{Mobility}(y)} = \frac{\sigma_{y''}/\sigma_{y'}}{\sigma_{y'}/\sigma_y} \tag{3.6}$$

where $\sigma_{y''}$ is the standard deviation of the second derivative of the signal. Because it is a ratio of two frequency-domain (Hz) equivalents, Complexity is a dimensionless scalar.

Together, these three parameters output exactly three scalar features per frequency band per channel, expanding the univariate feature space by explicitly encoding the temporal structural variations that basic amplitude statistics fundamentally miss.

3.7.4 Peak and Macro Frequency Extraction

Beyond traditional amplitude statistics, two specialized time-domain features were engineered to capture the dominant oscillatory and structural patterns within each specific frequency band of the 1.0-second analysis window. For a discrete band-pass filtered EEG signal vector $y(t)$ of length $N = 256$ and total duration $T = 1.0$ s, the features are calculated as follows:

Peak Frequency

Rather than extracting the maximum bin from a Fourier transform, Peak Frequency is calculated directly in the time domain by identifying the total number of local maxima (peaks) within the signal. A point $y(t)$ is considered a peak if it is strictly greater than its immediate neighbors. Let P be the set of all peak indices. The Peak Frequency f_{peak} is defined as the absolute peak count normalized by the window duration:

$$f_{peak} = \frac{|P|}{T} \tag{3.7}$$

The input dimension is the time-series amplitude vector, and the output is a scalar frequency representing the raw, jagged oscillatory rate of the frequency band in Hertz (Hz).

Macro Frequency

While Peak Frequency captures every local maximum, it is highly sensitive to noise and high-frequency jitter. To isolate the true, underlying macroscopic oscillatory bursts, a novel Macro Frequency algorithm was developed.

First, a dynamic amplitude threshold θ_{peak} is calculated as the mean amplitude of all previously identified peaks:

$$\theta_{peak} = \frac{1}{|P|} \sum_{p \in P} y(p) \tag{3.8}$$

The signal is then converted into a binary sequence $B(t)$ where $B(t) = 1$ if $y(t) \geq \theta_{peak}$ and 0 otherwise.

Because a single macroscopic wave might contain slight amplitude dips that prematurely break a contiguous block of ones, the algorithm calculates the average length of all zero-sequences (gaps), denoted as $\bar{G}$. Any gap in $B(t)$ with a length $g \leq \bar{G}$ is mathematically bridged (converted to ones), producing a smoothed binary sequence $B_{smooth}(t)$.

Finally, the Macro Frequency f_{macro} is defined as the total number of distinct, contiguous blocks of ones in $B_{smooth}(t)$, normalized by the window duration:

$$f_{macro} = \frac{\text{BlockCount}(B_{smooth})}{T} \tag{3.9}$$

The output is a stabilized, macroscopic frequency in Hertz (Hz) that robustly captures the count of dominant structural waves within the cognitive window, effectively ignoring micro-fluctuations.

3.8 Result Sythesis Pipeline

3.8.1 Temporal Cross-Validation and Data Balancing

Before proceeding to statistical evaluation and modeling, the inherent class imbalance depicted in Figure 3.17, where the `STAY` class (blue) heavily dominates the `SKIP` class (red), had to be addressed (executed via `05-12_data_analysis_and_results/scripts/step04_balance_split.py`). To prevent the models from trivially predicting the majority class, the `STAY` class was randomly undersampled for each participant to achieve a strict 50/50 balance, per participant, as depicted in Figure 3.18.

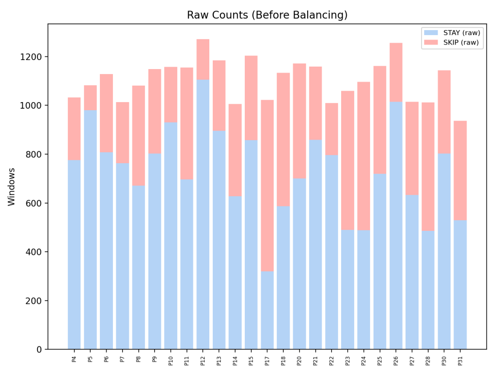

Figure 3.17: Raw window counts before balancing showing the extreme STAY dominance.

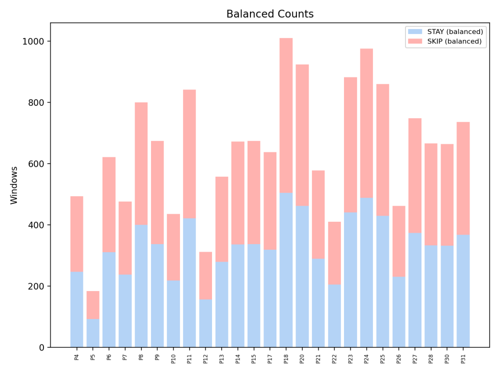

Figure 3.18: Window counts after undersampling, achieving a strict 50/50 ratio per participant.

To mitigate temporal leakage due to autocorrelation between adjacent EEG windows, blocked temporal 5-fold cross-validation was applied. Each participant's data were partitioned chronologically into five folds with equal counts of minority-class (`SKIP`) instances. Figure 3.19 shows minority-class occurrences (red) and fold boundaries (black), defined by equal splits of these events, per participant.

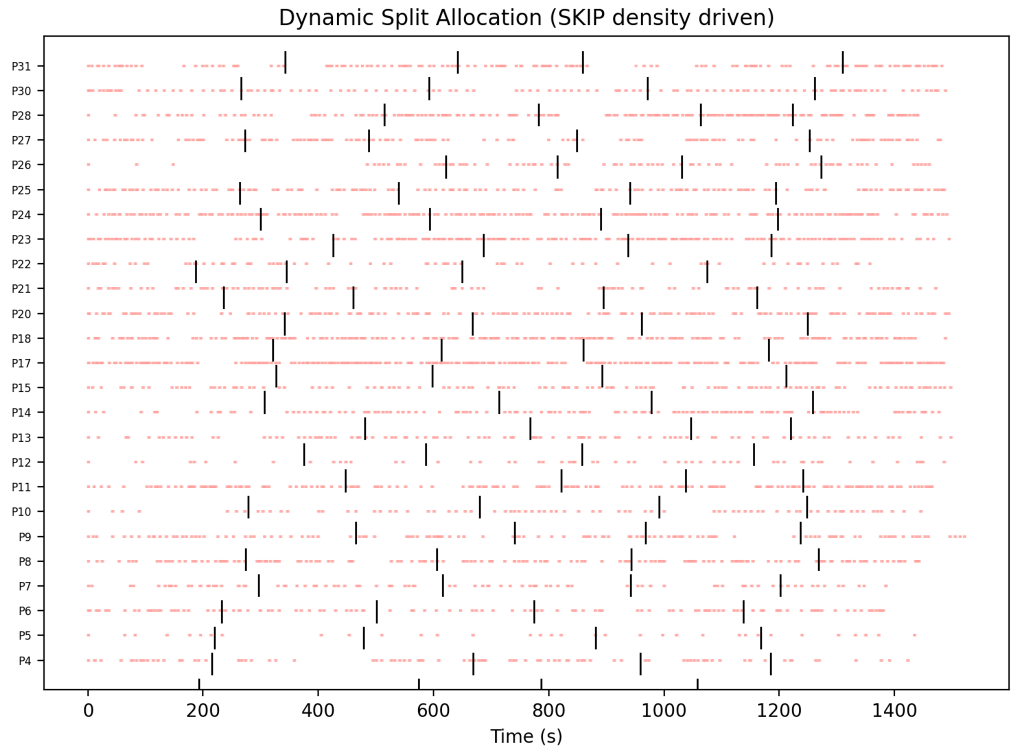

Figure 3.19: Each participant's data is temporally split into 5 folds containing equal amounts of the minority SKIP class.

To rigorously prevent temporal leakage, a minimum separation of 3 seconds between all training and testing windows was strictly enforced at every fold boundary. This constraint was implemented as a hard criterion: any violation automatically triggered pipeline termination, ensuring that no autocorrelated samples could enter the training or test sets. As verified in Figure 3.20, all fold boundary gaps (red points along the x-axis) exceed the predefined 3-second threshold (red dashed line), confirming compliance across the dataset. This procedure produces valid minority-balanced folds while preserving strict independence of the test data; an example for a single participant is shown in Figure 3.21, where the separation gaps between folds (colored blocks) are clearly visible.

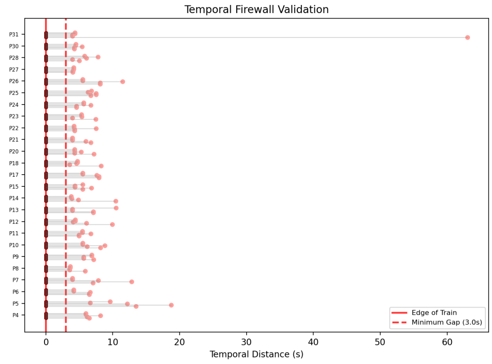

Figure 3.20: Temporal minimum required safety distance of 3 seconds between train and test windows is strictly fulfilled.

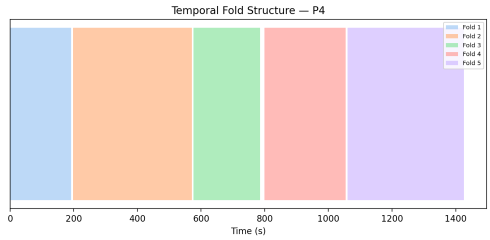

Figure 3.21: Temporal cross-validation split visualization, detailing the visible gaps between folds that successfully prevent data leakage.

3.8.2 Feature Separation Analysis

To mathematically evaluate the discriminative power of individual features prior to modeling, Cohen's d was computed to measure the standardized effect size between the `STAY` and `SKIP` distributions for each engineered feature (executed via `05-12_data_analysis_and_results/scripts/viz17_feature_ranking.py`).

Cohen's d is defined as:

$$d = \frac{\mu_{stay} - \mu_{skip}}{s_{pooled}} \tag{3.10}$$

where μ_{stay} and μ_{skip} represent the class-specific sample means, and s_{pooled} denotes the pooled standard deviation.

By calculating the absolute effect size ($|d|$), the pipeline systematically ranked all extracted features (e.g., `TP10_raw_std`) according to their individual ability to separate sustained cognitive engagement from the decision to disengage. Features demonstrating the highest absolute d values were designated as 'Top Features' and isolated for independent predictive baseline modeling.

3.8.3 Logistic Regression Modeling and Baselines

To benchmark the true predictive capability of the engineered EEG features against chance and clinical baselines, three distinct modeling approaches were evaluated on the temporally isolated test sets (executed via `05-12_data_analysis_and_results/scripts/step07_train_intra.py` and synthesized in `step16_parallel_universes.py`).

1. Coinflip Baseline: A uniform dummy classifier (`DummyClassifier(strategy='uniform')`) was utilized to establish an empirical chance baseline. Given the rigorous 50/50 balancing of the datasets, this approach reliably generated predictions hovering around the 50% chance level, verifying that no underlying structural bias existed within the cross-validation folds.

2. Engagement Index (EI) Baseline: The pre-calculated EI scalar (derived from $\beta/(\alpha+\theta)$) was scaled utilizing a standard scaler (subtracting the mean and scaling to unit variance) and fed as a singular feature into a Logistic Regression model (`LogisticRegression(class_weight='balanced')`). This established a clinical baseline, indicating how well classical engagement theory alone could predict the behavioral swipe decision in this specific short-form video context.

3. Top Feature Logistic Regression: To test the predictive power of individual statistical features independently, the features ranked highest via Cohen's d (e.g., specific band powers or statistical moments) were similarly scaled and evaluated using isolated Logistic Regression models. This isolated approach allows for direct comparison of individual feature efficacy without the confounding complexities of high-dimensional multivariate models.

3.8.4 Metrics and Significance Testing

To comprehensively evaluate model performance, standard classification metrics were computed for each cross-validation fold: Accuracy, Precision, Recall, and the F_1-score. In this context, correct prediction of the `STAY` class reflects successful detection of sustained cognitive engagement, whereas correct prediction of the `SKIP` class corresponds to identifying an impending disengagement decision within a 3–5 second pre-action window.

The F_1-score was used as the primary performance metric, as it balances Precision and Recall and is robust to class imbalance. Metrics were computed separately for each class by treating the respective class as the positive label (i.e., one-vs-rest evaluation for `SKIP` and `STAY`).

Metric definitions: Let TP, FP, TN, and FN denote true positives, false positives, true negatives, and false negatives, respectively.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Case 1: `SKIP` as the positive class

- **TP:** correctly predicted `SKIP` instances
- **FP:** `STAY` instances incorrectly predicted as `SKIP`
- **FN:** `SKIP` instances incorrectly predicted as `STAY`
- **TN:** correctly predicted `STAY` instances

Case 2: `STAY` as the positive class

- **TP:** correctly predicted `STAY` instances
- **FP:** `SKIP` instances incorrectly predicted as `STAY`
- **FN:** `STAY` instances incorrectly predicted as `SKIP`
- **TN:** correctly predicted `SKIP` instances

A class-specific F_1-score substantially exceeding chance level indicates that the model captures discriminative structure beyond random guessing for that class. For reference, under balanced random predictions, the expected F_1-score is approximately 0.5, whereas under strong class imbalance, this baseline decreases and should be interpreted relative to the class distribution.

To rigorously evaluate whether the Logistic Regression models utilizing individual EEG features outperformed the established baselines across the participant cohort, non-parametric statistical testing was employed. Specifically, the Wilcoxon signed-rank test was used to compute the statistical significance (p-value) of the performance differences between the paired model distributions (e.g., Top Feature model vs. EI Baseline model, or EI Baseline model vs. 50% chance level) across all participants (executed via `05-12_data_analysis_and_results/scripts/step17_thesis_figures_and_tables.py`).

The rank-biserial correlation (r) was also computed to quantify the effect size of the Wilcoxon test, indicating the strength of the model's superiority over the compared baseline. This structured chain of statistical proof robustly determines whether individual EEG features can provide a significant improvement over both pure chance and classical clinical engagement metrics when attempting to predict real-world short-form video interactions.

Chapter 4

Results and Discussion

This chapter presents the results of a multi-stage experimental investigation. This thesis delivers five core scientific contributions:

1. **Primary EEG Experiment under Naturalistic Conditions**

   The foundation of this work is the primary experiment (Section **??**), in which participants were recorded via EEG while engaging in naturalistic TikTok browsing. This data collection phase constituted a substantial portion of the overall effort, requiring careful experimental design, technical implementation, and data acquisition under ecologically valid conditions. The resulting dataset forms the basis for all subsequent analyses and should be understood as a central contribution of this thesis.

2. **Reusable EEG Processing and Analysis Pipeline**

   In addition to the experimental dataset, a further key contribution lies in the development of a comprehensive script library and analysis pipeline (Section **??**). This framework enables fully reproducible processing, from automated EEG recording on standard laptop hardware to feature extraction, class separation, and machine learning-based prediction. As such, it not only supports the present analysis but also provides a transferable methodological toolset that can be readily adapted to related experimental paradigms.

3. **Behavioral Burst Analysis**

   We continue by focusing on the behavioral burst analysis (Section 4.3). As no burst rejection was implemented—due to scope constraints outlined in Section 3.6.5—this component is treated as a secondary result rather than the central analytical pathway.

4. **Feature Discrimination and Predictive Modeling**

   The primary results emerge from a multi-layer analysis (Section **??**): Using the Cohen's d method described in Section 3.8.2, we identify features that most effectively differentiate between the SKIP and STAY classes (Section 4.4.1). Building on this, a set of machine learning models as described in Section 3.8.3 is applied to predict scrolling behavior based on these class distinctions, challenge the conventional Engagement-Index, and best-performing models and features are reported (Section 4.4.2).

   As described in Section 3.8.4, these results are carefully introduced along their statistical significance, a crucial measure considering the sample size of n=25 participants, in order to give result metrics such as the F1-score the appropriate context for meaningful interpretation. While this prediction task serves as an initial proof of concept, it already indicates which features may provide sufficient discriminatory power.

5. **Neuroscientific and Behavioral Interpretation**

   The results (Section 4.4.1) offer insights into the brain regions and frequency bands most predictive of this behavior, thereby informing both theoretical interpretation and potential practical applications. Therefore, in the final section (Section 4.5), these findings are interpreted in the context of neuroscience, psychology, and behavioral research, highlighting their implications and real-world relevance.

4.1 Key Findings from the Experimental Paradigm

4.2 Reusable Methodological Pipeline

4.3 Characterizing the Behavioral Burst Phenomenon

As established in the methodology (Section 3.6.5), a burst is defined as a sequence of rapidly successive skip events reflecting frantic swiping. To quantify this burst severity, overlapping `SKIP` windows were grouped into "burst blocks." The block size n denotes the number of consecutive rapid swipes before a participant pauses to engage. A block of $n = 1$ represents standard behavior, whereas larger values (e.g., $n = 3$) indicate repeated rapid skipping without meaningful engagement.

Figure 4.1 shows this behavior at the participant level, revealing a clear dichotomy: some participants exhibit frequent large burst blocks ("fast scrollers"), while others predominantly show $n = 1$ behavior ("careful scrollers"). Figure 4.2 aggregates these patterns, confirming that while $n = 1$ dominates, a long tail of burst behavior persists.

Finally, Figure 4.3 illustrates the sensitivity to the chosen window size by repeating the analysis with a ±5.0-second window (10.0-second merge threshold). The wider threshold captures a larger portion of the dwell time distribution and merges many isolated swipes into larger burst blocks, substantially reducing the proportion of $n = 1$ events. Specifically, under the ±3.0s window paradigm, 67.4% of all swipe events are isolated $n = 1$ sequences (participant-level average: 69.1% ± 16.2%). In contrast, widening the window to ±5.0s reduces the overall proportion of isolated swipes to 48.9% (participant-level average: 46.9% ± 20.8%). This underscores the strong dependence on parameter selection and the need for systematic optimization in future work.

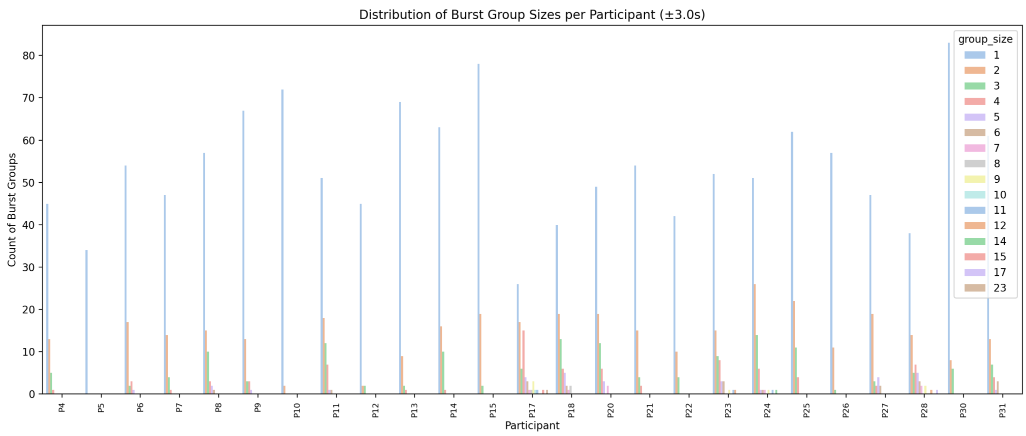

Figure 4.1: Distribution of burst group sizes per participant (using the ±3.0s window). This highlights the distinct behavioral difference between "frantic scrollers" (larger burst blocks) and "careful scrollers" (predominantly isolated $n = 1$ swipes).

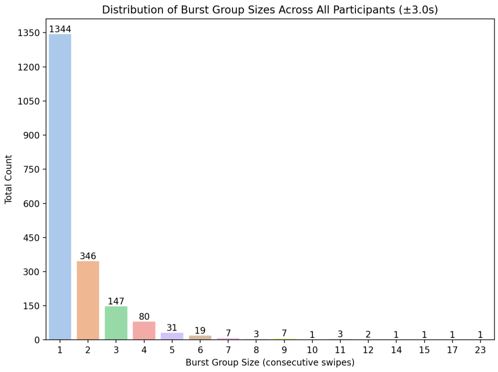

Figure 4.2: Total burst group sizes across all participants (±3.0s window).

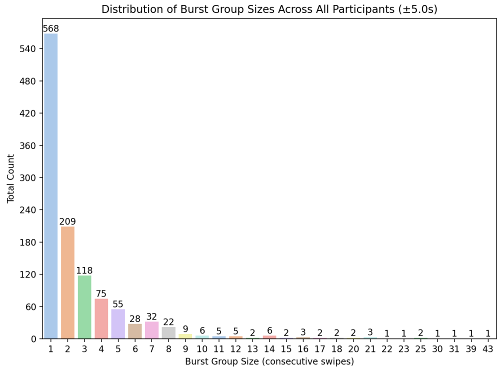

Figure 4.3: Total burst group sizes across all participants (±5.0s window). The wider threshold computationally merges many normal swipes into larger burst blocks.

4.4 Evaluating the Engagement Index

4.4.1 Feature Importance and Neurophysiological Correlates

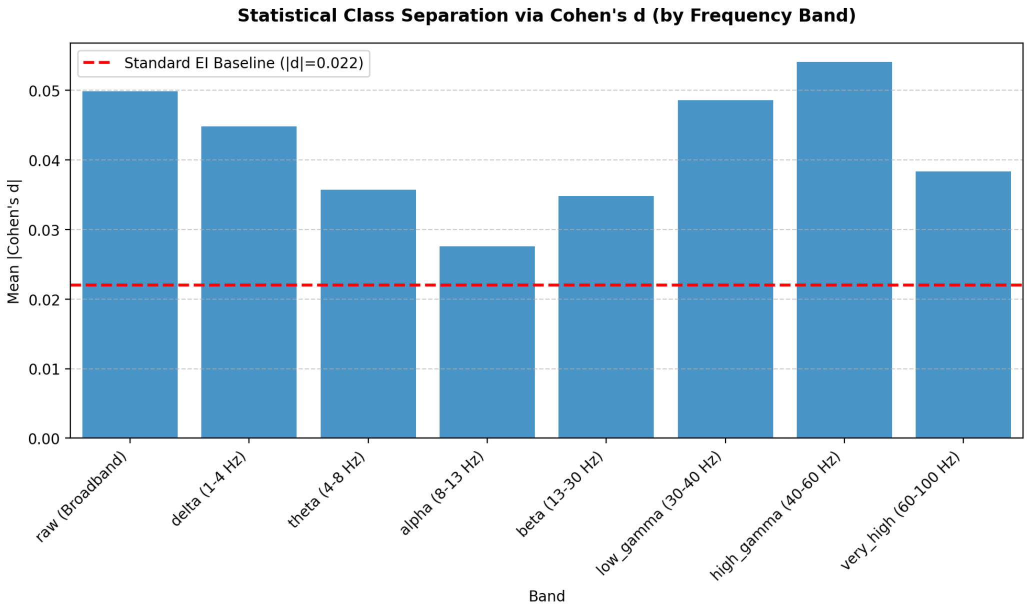

Figure 4.4: Cohen's d effect size by frequency band.

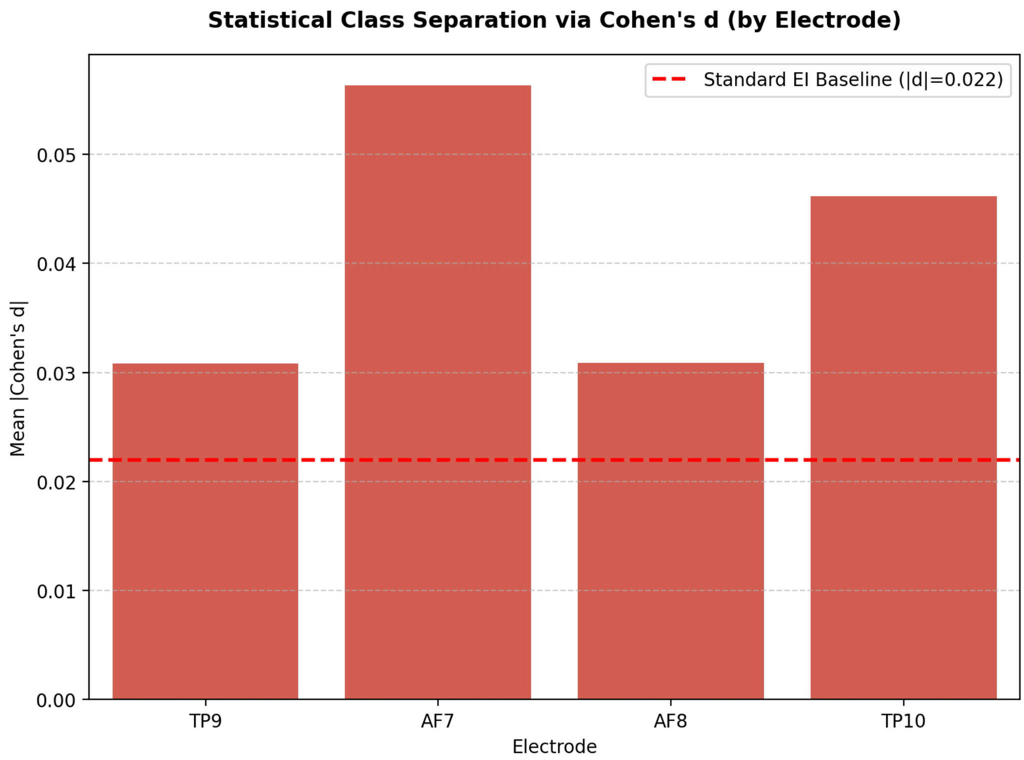

Figure 4.5: Cohen's d effect size by electrode.

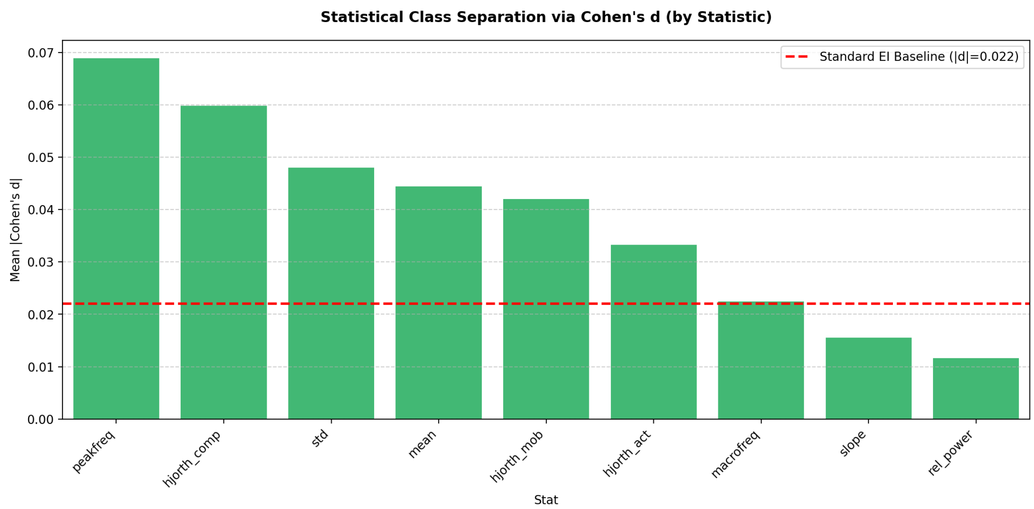

Figure 4.6: Cohen's d effect size by statistical moment.

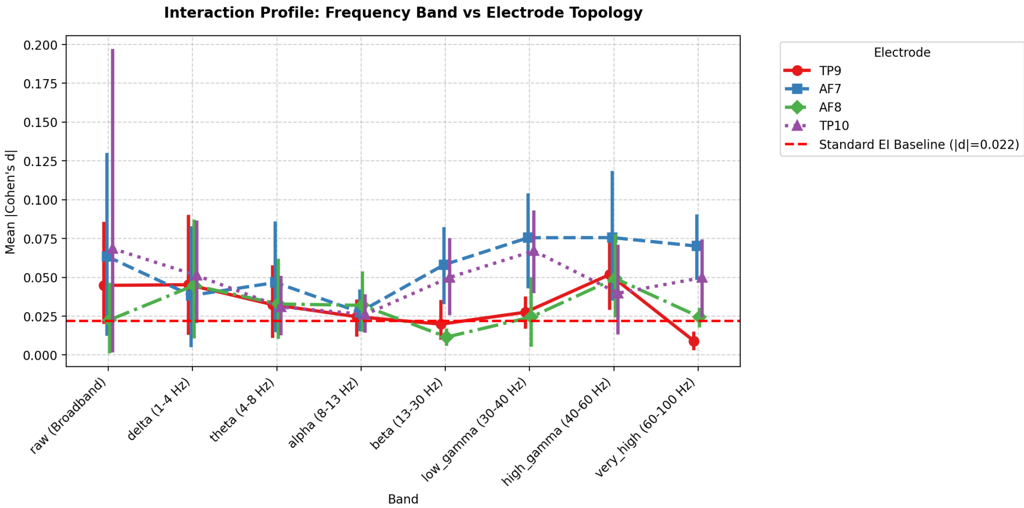

Figure 4.7: Cohen's d effect size interactions.

4.4.2 Model Performance and Benchmark Comparison

Inter Step 1: EI vs Coin Flip

Metric	Coin Flip (Value)	Coin Flip (Sig)	EI LR (Value)	EI LR (Sig)
F1 Train (STAY)	50.3%	-	62.0%	TRUE
F1 Test (STAY)	50.8%	-	62.5%	TRUE
Precision (STAY)	49.9%	-	49.4%	FALSE
Recall (STAY)	51.7%	-	89.2%	TRUE
F1 Train (SKIP)	49.6%	-	20.6%	TRUE
F1 Test (SKIP)	48.9%	-	10.7%	TRUE
Precision (SKIP)	49.9%	-	23.5%	TRUE
Recall (SKIP)	48.0%	-	9.6%	TRUE

Figure 4.8: Inter-subject performance metrics.

Metric	Coin Flip	EI LR	
	Val	Val	Sig
F1 Train (STAY)	50.3%	62.0%	TRUE
F1 Test (STAY)	50.8%	62.5%	TRUE
Precision (STAY)	49.9%	49.4%	FALSE
Recall (STAY)	51.7%	89.2%	TRUE
F1 Train (SKIP)	49.6%	20.6%	TRUE
F1 Test (SKIP)	48.9%	10.7%	TRUE
Precision (SKIP)	49.9%	23.5%	TRUE
Recall (SKIP)	48.0%	9.6%	TRUE

Table 4.1: Step 1: EI vs Coin Flip. Significance tested using Wilcoxon signed-rank test against Coin Flip (two-sided, $\alpha = 0.05$, N=25).

Intra Step 1: EI vs Coin Flip

Metric	Coin Flip (Value)	Coin Flip (Sig)	EI LR (Value)	EI LR (Sig)
F1 Train (STAY)	50.6%	-	53.8%	TRUE
F1 Test (STAY)	52.4%	-	50.9%	TRUE
Precision (STAY)	50.1%	-	49.1%	TRUE
Recall (STAY)	55.0%	-	52.9%	TRUE
F1 Train (SKIP)	49.4%	-	49.6%	FALSE
F1 Test (SKIP)	47.5%	-	46.9%	FALSE
Precision (SKIP)	50.1%	-	48.9%	TRUE
Recall (SKIP)	45.2%	-	45.1%	FALSE

Figure 4.9: Intra-subject performance metrics.

To rigorously determine whether a predictive model meaningfully outperforms the random baseline (Coin Flip) or the standard reference (Engage-

Metric	Coin Flip	EI LR	
	Val	Val	Sig
F1 Train (STAY)	50.6%	53.8%	TRUE
F1 Test (STAY)	52.4%	50.9%	TRUE
Precision (STAY)	50.1%	49.1%	TRUE
Recall (STAY)	55.0%	52.9%	TRUE
F1 Train (SKIP)	49.4%	49.6%	FALSE
F1 Test (SKIP)	47.5%	46.9%	FALSE
Precision (SKIP)	50.1%	48.9%	TRUE
Recall (SKIP)	45.2%	45.1%	FALSE

Table 4.2: Step 1: EI vs Coin Flip. Significance tested using Wilcoxon signed-rank test against Coin Flip (two-sided, $\alpha = 0.05$, N=25).

ment Index, EI), statistical significance was evaluated using the Wilcoxon signed-rank test. Unlike standard parametric tests (such as the paired t-test) which assume performance metrics are normally distributed, the Wilcoxon test is a non-parametric alternative. This makes it particularly robust for highly individual, small-sample physiological data such as EEG ($N = 25$ participants).

Rather than comparing a single aggregated mean across the dataset, the test utilizes paired metrics derived from the 25 cross-validation folds in a Leave-One-Group-Out (LOGO-CV) setup, where each fold trains a generalized model on 24 participants and evaluates it strictly on the held-out 25th participant. The procedure calculates the difference in performance (e.g., F1-score) between the evaluated model and the baseline for each specific participant. By ranking the absolute values of these differences and applying their original signs, the test evaluates whether the median of the differences is significantly shifted from zero. This participant-level comparison ensures that a model only achieves significance if it consistently outperforms the baseline across the majority of individuals, preventing a scenario where exceptional performance on a few participants artificially inflates the overall mean. A visual representation of these paired differences across all participants is provided in the significance heatmap (see Figure ??), which color-codes the exact performance gap for each participant.

The algorithm employs a two-sided Wilcoxon test with a standard significance threshold ($\alpha = 0.05$). Because a two-sided test detects significant differences in either direction (meaning the model could be significantly better *or* significantly worse than the baseline), a directional validation check is applied post-hoc. A model's performance is deemed statistically significantly superior only if it meets two strict conditions: first, the test must yield $p < 0.05$; second, the arithmetic mean of the model's scores across all participants must be strictly greater than the baseline's mean. The results of this rigorous evaluation, including the final validated significance flags for each metric, are summarized in Table 4.1. Furthermore, the test assumes that metric distributions are independent between participants, but appropriately treats models evaluated on the identical set of participant features as dependent, paired samples. Zero-differences (ties) are naturally handled by standard removal from the ranking process.

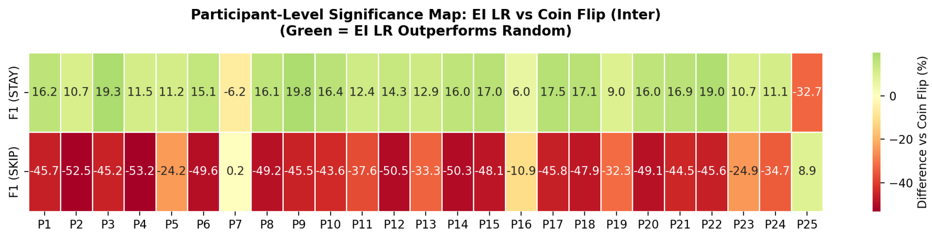

Figure 4.10: Inter-subject significance analysis.

To rigorously determine whether a predictive model meaningfully outperforms the random baseline (Coin Flip) or the standard reference (Engagement Index, EI), statistical significance was evaluated using the Wilcoxon signed-rank test. Unlike standard parametric tests (such as the paired t-test) which assume performance metrics are normally distributed, the Wilcoxon test is a non-parametric alternative. This makes it particularly robust for highly individual, small-sample physiological data such as EEG ($N = 25$ participants).

Rather than comparing a single aggregated mean across the dataset, the test utilizes paired metrics derived from the 25 participant-specific evaluations, where an individualized model is trained and tested exclusively on each participant's own data splits. The procedure calculates the difference in performance (e.g., F1-score) between the evaluated model and the baseline for each specific participant. By ranking the absolute values of these differences and applying their original signs, the test evaluates whether the median of the differences is significantly shifted from zero. This participant-level comparison ensures that a model only achieves significance if it consistently outperforms the baseline across the majority of individuals, preventing a scenario where exceptional performance on a few participants artificially inflates the overall mean. A visual representation of these paired differences across all participants is provided in the significance heatmap (see Figure **??**), which color-codes the exact performance gap for each participant.

The algorithm employs a two-sided Wilcoxon test with a standard significance threshold ($\alpha = 0.05$). Because a two-sided test detects significant differences in either direction (meaning the model could be significantly better *or* significantly worse than the baseline), a directional validation check is applied post-hoc. A model's performance is deemed statistically significantly superior only if it meets two strict conditions: first, the test must yield $p < 0.05$; second, the arithmetic mean of the model's scores across all participants must be strictly greater than the baseline's mean. The results of this rigorous evaluation, including the final validated significance flags for each metric, are summarized in Table 4.2. Furthermore, the test assumes that metric distributions are independent between participants, but appropriately treats models evaluated on the identical set of participant features as dependent, paired samples. Zero-differences (ties) are naturally handled by standard removal from the ranking process.

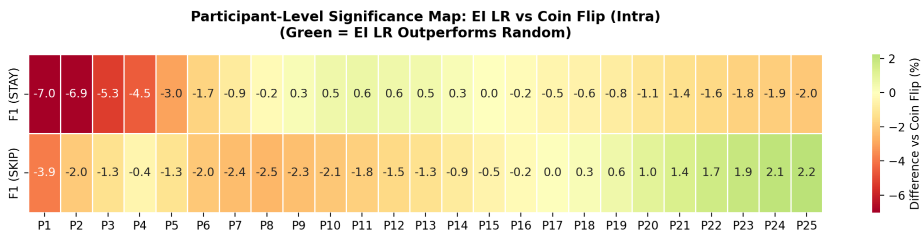

Figure 4.11: Intra-subject significance analysis.

4.5 Neuroscientific and Psychological Interpretation

4.6 Behavioral Burst Analysis (Detailed Results)

Methodology for Significance Testing

To rigorously determine whether a predictive model meaningfully outperforms a random baseline (Coin Flip) or a standard reference (Engagement Index, EI LR), this pipeline employs the Wilcoxon signed-rank test. Standard parametric tests assume that the performance metrics across participants are normally distributed. In small-sample physiological data (N=25 participants), this assumption is often violated. Performance is evaluated using paired metrics from the 25 individual cross-validation folds. The test calculates the difference in performance between the evaluated model and the baseline model for each participant. The algorithm employs a two-sided Wilcoxon test with a standard significance threshold ($\alpha = 0.05$). To strictly guarantee superiority, a directional check is applied post-hoc: the arithmetic mean of the model's scores across all participants must be strictly greater than the arithmetic mean of the baseline.

To quantify the theoretical separability of the extracted features prior to any model training, Cohen's d is utilized. As a standardized effect size, Cohen's d provides a mathematically pure measure of class separation that is invariant to the absolute scale or unit of the feature (e.g., microvolts versus power spectral density). It answers precisely how many standard deviations separate the centers of the two distinct classes.

The calculation operates on the window level, utilizing the entire global pool of validated EEG windows from all participants simultaneously. For any given engineered feature, the arithmetic mean is calculated across all windows labeled as STAY (μ_{stay}) and across all windows labeled as SKIP (μ_{skip}). Next, the variance within the STAY class (σ^2_{stay}) and the variance within the SKIP class (σ^2_{skip}) are calculated. These are averaged to compute the pooled variance, whose square root yields the pooled standard deviation (s_p), representing the average noise of the feature regardless of its class label:

$$s_p = \sqrt{\frac{\sigma^2_{stay} + \sigma^2_{skip}}{2}}$$

Finally, Cohen's d is calculated by taking the absolute difference between the two class means and dividing it by this pooled standard deviation:

$$|d| = \frac{|\mu_{stay} - \mu_{skip}|}{s_p}$$

By standardizing the difference in means against the inherent variance of the data, this calculation simultaneously rewards features where the class centers are far apart while heavily penalizing features that exhibit excessive noise or overlap. This allows features of entirely different natures to be ranked on a single, standardized axis of separability.

Applying this methodology to the full extracted feature set reveals clear frontrunners in class separability. The highest-ranking feature, TP10 raw std, achieves a separation of $|d| = 0.196$ standard deviations between the STAY and SKIP classes. This leading feature provides a separation magnitude that is 0.028 standard deviations wider than the second-best feature, AF7 high gamma mean ($|d| = 0.168$).

Furthermore, the second-best feature maintains a separation advantage of 0.023 standard deviations over the third-ranked feature, AF7 high gamma hjorth mob ($|d| = 0.145$). This objective standardization confirms that the uppermost features possess a notably distinct capacity to cleanly divide the dataset independent of any underlying machine learning classification bounds.

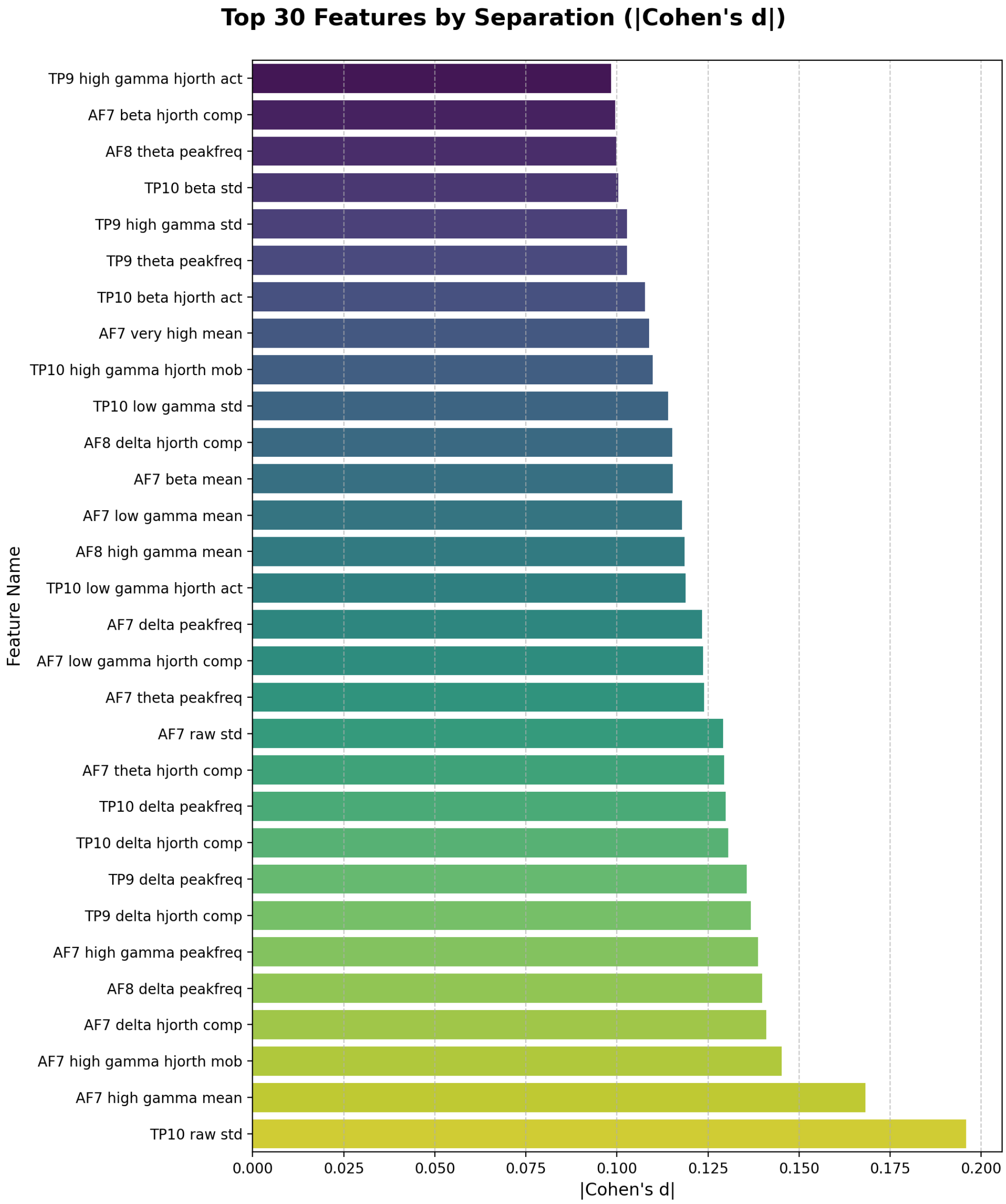

Figure 4.12: Top features ranked by Cohen's d.

Methodology:
To ensure a 1-to-1 Apples-to-Apples comparison with the 1D Engagement Index, each of the macro 'Top Parameters' (Band, Electrode, Stat) is represented by its strongest single constituent feature.

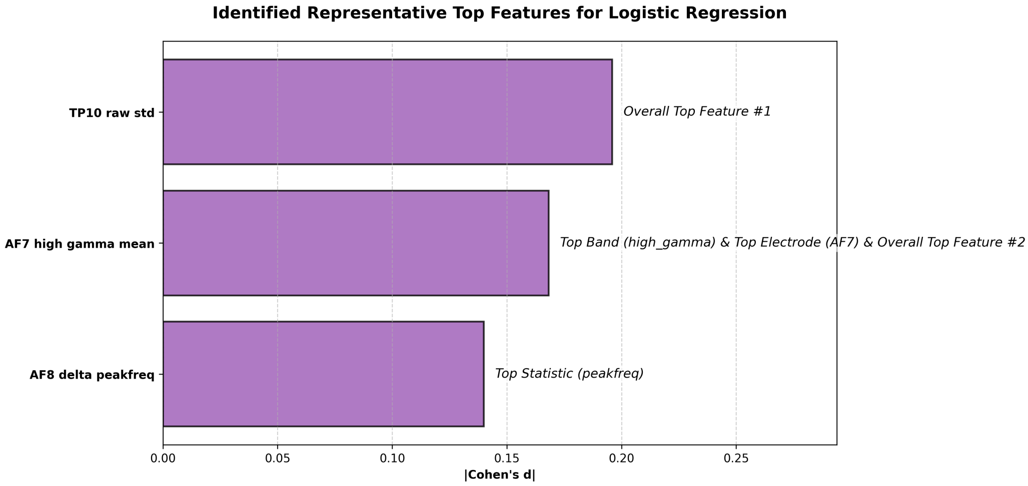

Figure 4.13: Identifying top predictive features.

Selected Features


```
{
    "AF7_high_gamma_mean": {
        "d": 0.16824331879615784,
        "labels": ["Top Band (high_gamma)", "Top Electrode (AF7)", "Overall Top
    },
    "AF8_delta_peakfreq": {
        "d": 0.13990937173366547,
        "labels": ["Top Statistic (peakfreq)"]
    },
    "TP10_raw_std": {
        "d": 0.1959162801504135,
        "labels": ["Overall Top Feature #1"]
    }
}
```

Inter Step 5: AF7 high gamma mean vs EI

Metric	EI LR (Value)	EI LR (Sig)	Proposed Feature (Value)	Proposed Feature (Sig)
F1 Train (STAY)	62.0%	-	54.3%	TRUE
F1 Test (STAY)	62.5%	-	45.5%	TRUE
Precision (STAY)	49.4%	-	44.3%	FALSE
Recall (STAY)	89.2%	-	58.1%	TRUE
F1 Train (SKIP)	20.6%	-	45.5%	TRUE
F1 Test (SKIP)	10.7%	-	35.5%	TRUE
Precision (SKIP)	23.5%	-	52.4%	TRUE
Recall (SKIP)	9.6%	-	42.7%	TRUE

Figure 4.14: Inter-subject performance using Feature 1.

Inter Step 5: AF8 delta peakfreq vs EI

Metric	EI LR (Value)	EI LR (Sig)	Proposed Feature (Value)	Proposed Feature (Sig)
F1 Train (STAY)	62.0%	-	57.4%	TRUE
F1 Test (STAY)	62.5%	-	48.3%	TRUE
Precision (STAY)	49.4%	-	52.8%	TRUE
Recall (STAY)	89.2%	-	54.6%	TRUE
F1 Train (SKIP)	20.6%	-	54.1%	TRUE
F1 Test (SKIP)	10.7%	-	45.5%	TRUE
Precision (SKIP)	23.5%	-	53.1%	TRUE
Recall (SKIP)	9.6%	-	48.7%	TRUE

Figure 4.15: Inter-subject performance using Feature 2.

Inter Step 5: TP10 raw std vs EI

Metric	EI LR (Value)	EI LR (Sig)	Proposed Feature (Value)	Proposed Feature (Sig)
F1 Train (STAY)	62.0%	-	49.6%	TRUE
F1 Test (STAY)	62.5%	-	46.0%	TRUE
Precision (STAY)	49.4%	-	47.5%	FALSE
Recall (STAY)	89.2%	-	51.0%	TRUE
F1 Train (SKIP)	20.6%	-	52.4%	TRUE
F1 Test (SKIP)	10.7%	-	44.3%	TRUE
Precision (SKIP)	23.5%	-	43.0%	TRUE
Recall (SKIP)	9.6%	-	50.8%	TRUE

Figure 4.16: Inter-subject performance using Feature 3.

Intra Step 5: AF7 high gamma mean vs EI

Metric	EI LR (Value)	EI LR (Sig)	Proposed Feature (Value)	Proposed Feature (Sig)
F1 Train (STAY)	53.8%	-	52.6%	TRUE
F1 Test (STAY)	50.9%	-	52.1%	FALSE
Precision (STAY)	49.1%	-	49.9%	TRUE
Recall (STAY)	52.9%	-	54.6%	FALSE
F1 Train (SKIP)	49.6%	-	52.9%	TRUE
F1 Test (SKIP)	46.9%	-	47.2%	FALSE
Precision (SKIP)	48.9%	-	49.8%	TRUE
Recall (SKIP)	45.1%	-	45.0%	FALSE

Figure 4.17: Intra-subject performance using Feature 1.

Intra Step 5: AF8 delta peakfreq vs EI

Metric	EI LR (Value)	EI LR (Sig)	Proposed Feature (Value)	Proposed Feature (Sig)
F1 Train (STAY)	53.8%	-	83.0%	TRUE
F1 Test (STAY)	50.9%	-	49.6%	TRUE
Precision (STAY)	49.1%	-	50.3%	TRUE
Recall (STAY)	52.9%	-	48.9%	TRUE
F1 Train (SKIP)	49.6%	-	83.1%	TRUE
F1 Test (SKIP)	46.9%	-	51.0%	TRUE
Precision (SKIP)	48.9%	-	50.3%	TRUE
Recall (SKIP)	45.1%	-	51.7%	TRUE

Figure 4.18: Intra-subject performance using Feature 2.

Intra Step 5: TP10 raw std vs EI

Metric	EI LR (Value)	EI LR (Sig)	Proposed Feature (Value)	Proposed Feature (Sig)
F1 Train (STAY)	53.8%	-	52.4%	TRUE
F1 Test (STAY)	50.9%	-	49.3%	TRUE
Precision (STAY)	49.1%	-	48.2%	TRUE
Recall (STAY)	52.9%	-	50.4%	TRUE
F1 Train (SKIP)	49.6%	-	52.6%	TRUE
F1 Test (SKIP)	46.9%	-	46.8%	FALSE
Precision (SKIP)	48.9%	-	47.9%	TRUE
Recall (SKIP)	45.1%	-	45.7%	FALSE

Figure 4.19: Intra-subject performance using Feature 3.

Diagnostic Report (Inter)

Model: EI LR

Train F1 = 62.0%, Test F1 = 62.5%, Precision = 49.4%, Recall = 89.2%.
Significance vs Coin Flip: p >= 0.05
Diagnosis: Random chance performance (No signal). The F1 score is not statistically significantly better than the Coin Flip baseline (p >= 0.05).

Model: TP10 raw std LR

Train F1 = 49.6%, Test F1 = 46.0%, Precision = 47.5%, Recall = 51.0%.
Significance vs Coin Flip: p < 0.05
Diagnosis: Genuine predictive power. Healthy generalization with statistically identical train and test performance.

Model: AF7 high gamma mean LR

Train F1 = 54.3%, Test F1 = 45.5%, Precision = 44.3%, Recall = 58.1%.
Significance vs Coin Flip: p < 0.05
Diagnosis: Mild overfit. The model demonstrates genuine predictive power with expected minor performance degradation on unseen data.

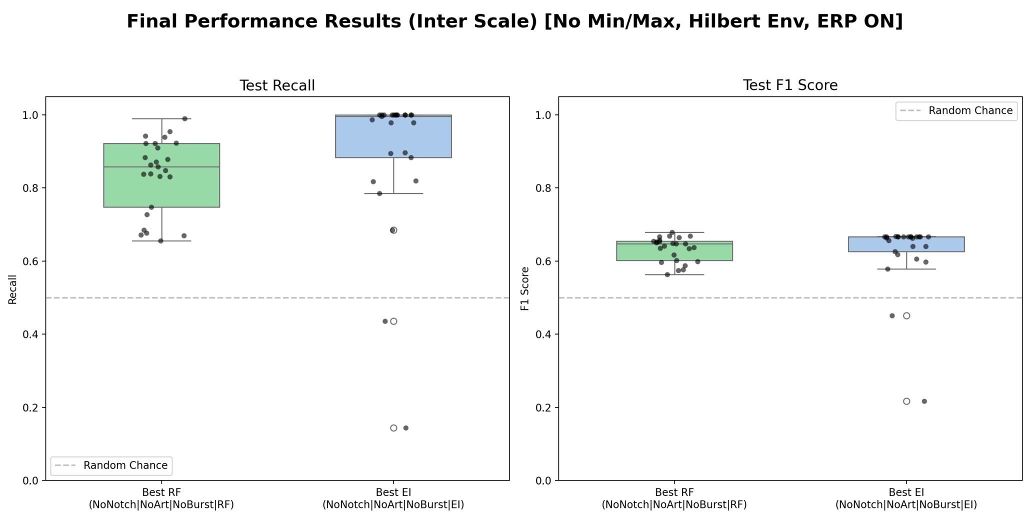

Figure 4.20: Final inter-subject performance summary.

Inter Model Metrics (Wilcoxon signed-rank test, N=25)

Metric	Coin Flip (Value)	Coin Flip (Sig)	EI LR (Value)	EI LR (Sig)	TP10 raw std LR (Value)	TP10 raw std LR (Sig)	AF7 high gamma mean LR (Value)	AF7 high gamma mean LR (Sig)
F1 Train (STAY)	50.3%	-	62.0%	-	49.6%	TRUE	54.3%	TRUE
F1 Test (STAY)	50.8%	-	62.5%	-	46.0%	TRUE	45.5%	TRUE
Precision (STAY)	49.9%	-	49.4%	-	47.5%	FALSE	44.3%	FALSE
Recall (STAY)	51.7%	-	89.2%	-	51.0%	TRUE	58.1%	TRUE
F1 Train (SKIP)	49.6%	-	20.6%	-	52.4%	TRUE	45.5%	TRUE
F1 Test (SKIP)	48.9%	-	10.7%	-	44.3%	FALSE	35.5%	FALSE
Precision (SKIP)	49.9%	-	23.5%	-	43.0%	FALSE	52.4%	FALSE
Recall (SKIP)	48.0%	-	9.6%	-	50.8%	FALSE	42.7%	FALSE

Figure 4.21: Inter-subject metrics table ablation.

Inter Optimal Preprocessing Parameters

Parameter	EI LR	TP10 raw std LR	AF7 high gamma mean LR
Scale	Inter	-	-
Notch	NoNotch	-	-
Artifact	NoArt	-	-
Burst	NoBurst	-	-

Figure 4.22: Inter-subject optimal parameters ablation.

Diagnostic Report (Intra)

Model: EI LR

Train F1 = 53.8%, Test F1 = 50.9%, Precision = 49.1%, Recall = 52.9%.
Significance vs Coin Flip: $p >= 0.05$
Diagnosis: Random chance performance (No signal). The F1 score is not statistically significantly better than the Coin Flip baseline ($p >= 0.05$).

Model: TP10 raw std LR

Train F1 = 52.4%, Test F1 = 49.3%, Precision = 48.2%, Recall = 50.4%.
Significance vs Coin Flip: $p < 0.05$
Diagnosis: Genuine predictive power. Healthy generalization with statistically identical train and test performance.

Model: AF7 high gamma mean LR

Train F1 = 52.6%, Test F1 = 52.1%, Precision = 49.9%, Recall = 54.6%.
Significance vs Coin Flip: p >= 0.05
Diagnosis: Random chance performance (No signal). The F1 score is not statistically significantly better than the Coin Flip baseline (p >= 0.05).

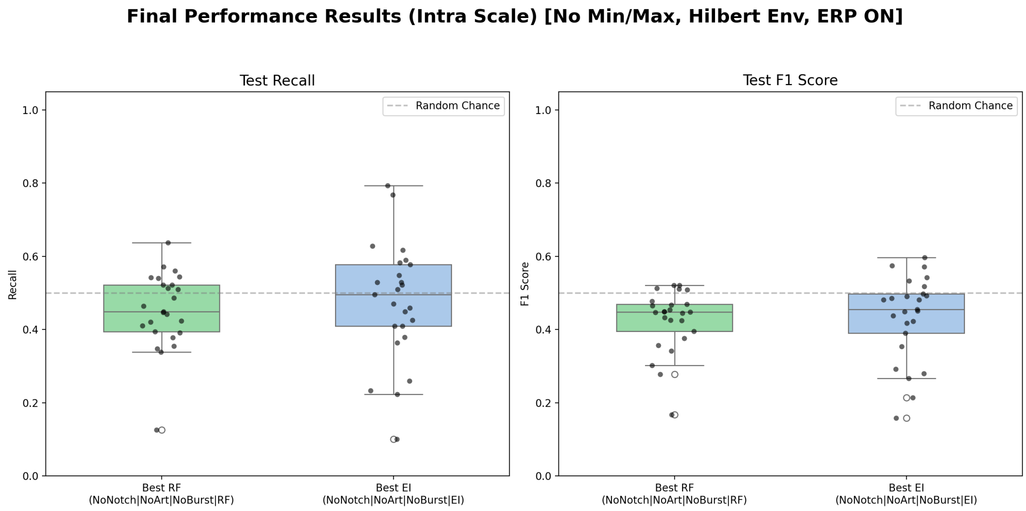

Figure 4.23: Final intra-subject performance summary.

Intra Model Metrics (Wilcoxon signed-rank test, N=25)

Metric	Coin Flip (Value)	Coin Flip (Sig)	EI LR (Value)	EI LR (Sig)	TP10 raw std LR (Value)	TP10 raw std LR (Sig)	AF7 high gamma mean LR (Value)	AF7 high gamma mean LR (Sig)
F1 Train (STAY)	50.6%	-	53.8%	-	52.4%	TRUE	52.6%	TRUE
F1 Test (STAY)	52.4%	-	50.9%	-	49.3%	TRUE	52.1%	TRUE
Precision (STAY)	50.1%	-	49.1%	-	48.2%	TRUE	49.9%	FALSE
Recall (STAY)	55.0%	-	52.9%	-	50.4%	TRUE	54.6%	TRUE
F1 Train (SKIP)	49.4%	-	49.6%	-	52.6%	TRUE	52.9%	TRUE
F1 Test (SKIP)	47.5%	-	46.9%	-	46.8%	FALSE	47.2%	FALSE
Precision (SKIP)	50.1%	-	48.9%	-	47.9%	TRUE	49.8%	FALSE
Recall (SKIP)	45.2%	-	45.1%	-	45.7%	FALSE	45.0%	FALSE

Figure 4.24: Intra-subject metrics table ablation.

Intra Optimal Preprocessing Parameters

Parameter	EI LR	TP10 raw std LR	AF7 high gamma mean LR
Scale	Intra	-	-
Notch	NoNotch	-	-
Artifact	NoArt	-	-
Burst	NoBurst	-	-

Figure 4.25: Intra-subject optimal parameters ablation.

4.7 Summary of Results

Does it answer the research question? Yes, with appropriate calibration. Within this sample, consumer-grade EEG can predict sustained engagement significantly better than random chance at the intra-subject level (53.8%, $p = 0.0015$ via blocked temporal 5-fold CV), but the effect is small ($\sim$3.8pp above chance) and individually variable. Cross-subject generalization fails completely (LOGO-CV: 49.2%). Simple clinical formulas (EI: 52.3%) are insufficient. Feature importance is broadly distributed across all frequency bands, with the model exploiting the full spectral structure rather than any single dominant band. The top individual predictors are high-gamma peak values at frontal sites, but no band achieves statistically significant dominance at the aggregate level.

4.7.1 9. Outlook and Implications for Future BCI Design

A critical implication of this thesis is the rejection of the "Universal BCI Equation" paradigm. The LOGO-CV validation directly demonstrates this: the Random Forest—a non-linear, 112-component model—completely fails to generalize across participants (49.2% accuracy), providing evidence that the bottleneck is biological idiosyncrasy, not mathematical complexity. Future development of consumer BCI systems for algorithmic media interaction should pivot toward personalized, high-dimensional machine learning calibration architectures. Additionally, this thesis demonstrates that standard overlapping-window evaluation in BCI research can inflate reported performance by $\sim$8 percentage points, highlighting the importance of temporally-separated evaluation protocols in future work.

Rank	Feature Name	$|$Cohen's $d|$
1	TP10 raw std	0.196
2	AF7 high gamma mean	0.168
3	AF7 high gamma hjorth mob	0.145
4	AF7 delta hjorth comp	0.141
5	AF8 delta peakfreq	0.140
6	AF7 high gamma peakfreq	0.139
7	TP9 delta hjorth comp	0.137
8	TP9 delta peakfreq	0.136
9	TP10 delta hjorth comp	0.131
10	TP10 delta peakfreq	0.130
11	AF7 theta hjorth comp	0.129
12	AF7 raw std	0.129
13	AF7 theta peakfreq	0.124
14	AF7 low gamma hjorth comp	0.124
15	AF7 delta peakfreq	0.123
16	TP10 low gamma hjorth act	0.119
17	AF8 high gamma mean	0.119
18	AF7 low gamma mean	0.118
19	AF7 beta mean	0.115
20	AF8 delta hjorth comp	0.115
21	TP10 low gamma std	0.114
22	TP10 high gamma hjorth mob	0.110
23	AF7 very high mean	0.109
24	TP10 beta hjorth act	0.108
25	TP9 theta peakfreq	0.103
26	TP9 high gamma std	0.103
27	TP10 beta std	0.100
28	AF8 theta peakfreq	0.100
29	AF7 beta hjorth comp	0.100
30	TP9 high gamma hjorth act	0.098

Table 4.3: Top 30 Features by Separation ($|$Cohen's $d|$).

Metric	EI LR	AF7 high gamma mean	
	Val	Val	Sig
F1 Train (STAY)	62.0%	54.3%	TRUE
F1 Test (STAY)	62.5%	45.5%	TRUE
Precision (STAY)	49.4%	44.3%	FALSE
Recall (STAY)	89.2%	58.1%	TRUE
F1 Train (SKIP)	20.6%	45.5%	TRUE
F1 Test (SKIP)	10.7%	35.5%	TRUE
Precision (SKIP)	23.5%	52.4%	TRUE
Recall (SKIP)	9.6%	42.7%	TRUE

Table 4.4: Step 5 (Inter): Proposed Feature (AF7 high gamma mean) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, N=25).

Metric	EI LR	AF8 delta peakfreq	
	Val	Val	Sig
F1 Train (STAY)	62.0%	57.4%	TRUE
F1 Test (STAY)	62.5%	48.3%	TRUE
Precision (STAY)	49.4%	52.8%	TRUE
Recall (STAY)	89.2%	54.6%	TRUE
F1 Train (SKIP)	20.6%	54.1%	TRUE
F1 Test (SKIP)	10.7%	45.5%	TRUE
Precision (SKIP)	23.5%	53.1%	TRUE
Recall (SKIP)	9.6%	48.7%	TRUE

Table 4.5: Step 5 (Inter): Proposed Feature (AF8 delta peakfreq) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, N=25).

Metric	EI LR	TP10 raw std	
	Val	Val	Sig
F1 Train (STAY)	62.0%	49.6%	TRUE
F1 Test (STAY)	62.5%	46.0%	TRUE
Precision (STAY)	49.4%	47.5%	FALSE
Recall (STAY)	89.2%	51.0%	TRUE
F1 Train (SKIP)	20.6%	52.4%	TRUE
F1 Test (SKIP)	10.7%	44.3%	TRUE
Precision (SKIP)	23.5%	43.0%	TRUE
Recall (SKIP)	9.6%	50.8%	TRUE

Table 4.6: Step 5 (Inter): Proposed Feature (TP10 raw std) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, N=25).

Metric	EI LR	AF7 high gamma mean	
	Val	Val	Sig
F1 Train (STAY)	53.8%	52.6%	TRUE
F1 Test (STAY)	50.9%	52.1%	FALSE
Precision (STAY)	49.1%	49.9%	TRUE
Recall (STAY)	52.9%	54.6%	FALSE
F1 Train (SKIP)	49.6%	52.9%	TRUE
F1 Test (SKIP)	46.9%	47.2%	FALSE
Precision (SKIP)	48.9%	49.8%	TRUE
Recall (SKIP)	45.1%	45.0%	FALSE

Table 4.7: Step 5 (Intra): Proposed Feature (AF7 high gamma mean) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, N=25).

Metric	EI LR	AF8 delta peakfreq	
	Val	Val	Sig
F1 Train (STAY)	53.8%	83.0%	TRUE
F1 Test (STAY)	50.9%	49.6%	TRUE
Precision (STAY)	49.1%	50.3%	TRUE
Recall (STAY)	52.9%	48.9%	TRUE
F1 Train (SKIP)	49.6%	83.1%	TRUE
F1 Test (SKIP)	46.9%	51.0%	TRUE
Precision (SKIP)	48.9%	50.3%	TRUE
Recall (SKIP)	45.1%	51.7%	TRUE

Table 4.8: Step 5 (Intra): Proposed Feature (AF8 delta peakfreq) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, N=25).

Metric	EI LR	TP10 raw std	
	Val	Val	Sig
F1 Train (STAY)	53.8%	52.4%	TRUE
F1 Test (STAY)	50.9%	49.3%	TRUE
Precision (STAY)	49.1%	48.2%	TRUE
Recall (STAY)	52.9%	50.4%	TRUE
F1 Train (SKIP)	49.6%	52.6%	TRUE
F1 Test (SKIP)	46.9%	46.8%	FALSE
Precision (SKIP)	48.9%	47.9%	TRUE
Recall (SKIP)	45.1%	45.7%	FALSE

Table 4.9: Step 5 (Intra): Proposed Feature (TP10 raw std) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, N=25).

Metric	Coin Flip	EI LR	TP10 raw std LR		AF7 high gamma mean LR	
	Val	Val	Val	Sig	Val	Sig
F1 Train (STAY)	50.3%	62.0%	49.6%	TRUE	54.3%	TRUE
F1 Test (STAY)	50.8%	62.5%	46.0%	TRUE	45.5%	TRUE
Precision (STAY)	49.9%	49.4%	47.5%	FALSE	44.3%	FALSE
Recall (STAY)	51.7%	89.2%	51.0%	TRUE	58.1%	TRUE
F1 Train (SKIP)	49.6%	20.6%	52.4%	TRUE	45.5%	TRUE
F1 Test (SKIP)	48.9%	10.7%	44.3%	FALSE	35.5%	FALSE
Precision (SKIP)	49.9%	23.5%	43.0%	FALSE	52.4%	FALSE
Recall (SKIP)	48.0%	9.6%	50.8%	FALSE	42.7%	FALSE

Table 4.10: Minimal Metrics for Inter Models. Significance tested using Wilcoxon signed-rank test against Standard EI (two-sided, $\alpha = 0.05$, N=25).

Model	F1 Test (STAY)	F1 Test (SKIP)

Table 4.11: Models significantly outperforming baseline on F1 Test (Inter)

Metric	Coin Flip	EI LR	TP10 raw std LR		AF7 high gamma mean LR	
	Val	Val	Val	Sig	Val	Sig
F1 Train (STAY)	50.6%	53.8%	52.4%	TRUE	52.6%	TRUE
F1 Test (STAY)	52.4%	50.9%	49.3%	TRUE	52.1%	TRUE
Precision (STAY)	50.1%	49.1%	48.2%	TRUE	49.9%	FALSE
Recall (STAY)	55.0%	52.9%	50.4%	TRUE	54.6%	TRUE
F1 Train (SKIP)	49.4%	49.6%	52.6%	TRUE	52.9%	TRUE
F1 Test (SKIP)	47.5%	46.9%	46.8%	FALSE	47.2%	FALSE
Precision (SKIP)	50.1%	48.9%	47.9%	TRUE	49.8%	FALSE
Recall (SKIP)	45.2%	45.1%	45.7%	FALSE	45.0%	FALSE

Table 4.12: Minimal Metrics for Intra Models. Significance tested using Wilcoxon signed-rank test against Standard EI (two-sided, $\alpha = 0.05$, N=25).

Model	F1 Test (STAY)	F1 Test (SKIP)

Table 4.13: Models significantly outperforming baseline on F1 Test (Intra)

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